ROOTECH

Energy Solution for
Human & Nature

ACCURA 3300E

High Accuracy Digital Power Meter

Supporting RS-485 Interface / Modbus RTU Protocol

User Guide [English]

Revision 2.02 October 14, 2024

LISTED

Notice

Symbols

Caution

Indicates the presence of dangerous voltage which can cause severe injury or death if proper precautions are not followed.

Caution

Alerts users of the presence of hazards which can cause injury to persons, property damage or damage to the device if proper precautions are not followed.

Note

Indicates major instructions for the installation, operation, and maintenance of the product.

Indicates alternative voltage or current.

Indicates direct voltage or current.

Installation Considerations

The installation, operation, and maintenance of the product should be performed only by qualified, competent personnel that have received appropriate training for high voltage and current devices.

Caution

If hazardous voltages are mishandled while installing and operating the product in the field, it can cause serious injury or death.

- During normal operation of the product, hazardous voltages are present on the terminals that connect PTs/CTs, digital inputs, power, and external I/O circuits. The PT/CT secondary can produce lethal voltage or current caused by the energy of the primary.
- Standard safety precautions should be followed during the installation and maintenance of the product, including removing PT fuses and shorting CT secondaries.
- Install the product in enclosures or a similar cabinet to prevent access to the terminal strips of the product after wiring.

Caution

Observe the following instructions, or it may cause permanent damage to the product.

To ensure proper device operation, make sure the following:

- The device is properly installed.
- The power specifications indicated on the product are met.

About the Manual

Rootech Inc. reserves the right to make changes in the product specifications shown in this User Guide without prior notice. We recommend that customers should obtain the latest information on product specifications and the manual before placing orders.

In the absence of written agreements, Rootech Inc. assumes no liability for applications assistance, customer's system design, or infringement of patents or copyrights induced by third parties arising from the use of the products described herein.

We endeavor to provide accurate information in the document. However, we are not responsible for any errors which may appear here and reserves the right to make changes without prior notice.

Limitation of Liability

Where applicable law allows and does not prohibit or restrict such limitation, Rootech, Inc's liability for this product shall be limited to the amount actually paid by customers for the product.

Warranty

Rootech Inc. provides a warranty for two years from the product receipt date, applicable only to the original purchaser of the products and software sold or licensed by the company.

To obtain warranty service, the purchased products should be free from serious defects in material and workmanship.

Software products are provided in an up-to-date version, therefore, we do not offer any warranty for software product.

In order to make a product claim under the warranties described above, the original purchaser should immediately contact Rootech headquarters. When we receive a claim, we will inspect the product either at the purchaser's location or at the company's facility after receiving the product from the customer. The customer is responsible for the cost of shipping the product. After inspecting the product, we will provide repair or replacement services for the product.

Rootech Inc. reserves the right to determine, at its discretion, whether to repair or replace the product or issue a refund for the product when the warranty period has expired or issues are not covered by warranty terms and conditions.

Limitation of Warranty

The warranty does not apply to the uninterrupted or error-free operation of the product and does not cover normal wear and tear. It also does not cover costs associated with the removal, installation, or troubleshooting of the customer's electrical systems.

Defects caused by any of the following factors are not covered under the warranty.

- Improper use(alteration, accident, misuse, abuse) or failure to act in accordance with installation, operation or maintenance instructions
- Unauthorized modifications, changes or repair attempts
- Failure to comply with applicable safety standards and regulations
- Damages incurred during transportation or storage of the product
- Damages caused by the force majeure event, including fire, flood, earthquakes, storm damage, overvoltage and lightning
- Defacing, altering, or removal of the original identification marks(trademark, serial number, etc.)

Rootech Inc. shall not be liable for any claims other than those made by customers, including the original purchaser, their employees, agents, or contractors for any loss, damage, or expense incurred related to the purchased product when the above warranty terms and conditions are not fulfilled.

Any technical assistance provided by Rootech's personnel or representatives in system design shall be considered a suggestion. The original purchaser is responsible for determining the feasibility of such suggestions and should conduct tests to verify their feasibility

It is the original purchaser's responsibility to determine the suitability of any product and associated documentation. The original purchaser should acknowledge that 100% uptime is not achievable due to possible hardware or software defects. The original purchaser should also recognize that such defects and failures may cause inaccuracies or malfunctions.

No agency, corporate entity, or employee of Rootech Inc. or any other companies has the authority to amend, modify or extend the terms and conditions of this warranty in any way without express written authorization from Rootech, Inc.

Standards Compliance

LISTED

Process Control Equipment, Electrical E324900

R-R-RTE-Accura3300E

Revision History

The following versions of the Accura 3300E User Guide have been released.

Revision	Date	Description
Revision 1.00	December 4, 2015	Initial release
Revision 1.01	February 20, 2016	The following wording "the description of the temperature" is changed to the temperature of Accura 3300E.
Revision 1.02	September 20, 2017	Changed UL certification code
Revision 1.03	January 2, 2018	Changed information on the Notice and Warranty sections
Revision 1.10	January 22, 2019	Specified Accura 3300E-5A and Accura 3300E-1A models
Revision 1.30	April 19, 2022	Changed the measurement accuracy of power quality Changed the color of the LCD backlight Updated the standards compliance section Added the number of devices operating at 115,200 bps Changed wiring diagrams
Revision 2.00	April 23, 2024	Changed the number of samples per cycle for voltage and current to 512 and 128 Changed available harmonic data to harmonics up to the 50th order Changed the frequency measuring range to 65 Hz Removed the descriptions of Aggregation Added IP54 specifications Added the image and description related to RS-485 shielded cables
Revision 2.01	July 19, 2024	Changed UL certification code Changed voltage and current accuracy notation in Appendix C
Revision 2.02	October 14, 2024	Added option information for the product: Accura 3300E-5A-AC220V/DC110V and Accura 3300E-1A-AC220V/DC110V

Contents

Figures

Chapter 1 Introduction

Overview

Real-Time Power Management in Switchgear Panels

Due to problems with the power supply, power facilities, and loads, various power quality problems frequently occur in switchgear panels. In this regard, energy managers have to implement proper preventive or follow-up measures by figuring out the cause of problems based on real-time power analysis.

Gapless Measurements of Voltage and Current: Average Values for 1-Second Intervals Available

Accura 3300E takes 512 voltage and 128 current samples per cycle and continuously provides average values for 1-second intervals, enabling real-time power analysis.

Accurate Measurements of Voltage, Current and Active Energy: ±0.2 % of Reading / IEC 62053-22 Class 0.5S Accuracy

Due to greenhouse gas emission regulations, it has recently become essential to establish enterprise-wide energy management systems for efficient energy usage at sites such as plants, manufacturing factories, and buildings. A key factor determining the reliability of energy management systems is the measurement accuracy of meters. Accura 3300E performs accurate measurements with ±0.2% of reading accuracy for voltage and current and meets IEC 62053-22 Class 0.5S standards for power and energy. Thus, this allows for accurate analysis and diagnosis of various problems in energy management at power facilities.

A Variety of Power Quality Data Available

Accura 3300E provides various power quality information including harmonics(up to the 50th order), crest factor, K-factor, and unbalance. This makes it possible for energy managers to easily identify changes in power quality at power facilities.

Proven Safety and Reliability with CE/UL/KC Certifications

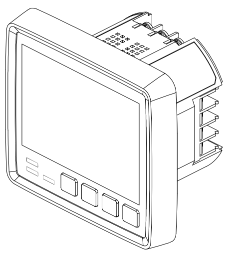
The internal and external structures (product and circuit design) of Accura 3300E meet safety and reliability requirements for CE(EN 55011: 2016/A11:2020, EN IEC 61326-1:2021, EN IEC 61326-2-1:2021, EN IEC 61000-3-2:2019/A1:2021, EN 61000-3-3:2013/A2:2021), UL(UL 61010-1 3rd edition, IEC 61010-2-030 2nd edition), and KC(EN 55011: 2016/A11:2020, EN IEC 61326-1:2021, EN IEC 61326-2-1:2021) standards.

Product Information

Product Features

Accura 3300E Digital Power Meter

Accura 3300E has ±0.2 % of reading accuracy for voltage and current measurements and meets IEC 62053-22 Class 0.5S standards for power and energy measurements. It also provides various information including harmonics(up to the 50th order), Crest factor, K-factor, and unbalance. which are essential for power quality management of switchgear panels.

Accura 3300E	Functions	
	**Display**	
	Shows real-time measurement data on the LCD screen	
	Shows measured values using bar graphs	
	General	
	Samples/cycle	Voltage: 512 samples/cycle
		Current: 128 samples/cycle
	Frequency measuring range	42 – 65 Hz
	True RMS measurement	Voltage/Current: ±0.2% of reading
	Residual voltage [1] , Residual current [2]	
	Demand, peak demand, predicted demand [3, 4]	
	Maximum and minimum values	
	Energy	
	Received/Delivered energy	IEC 62053-22 Class 0.5S
	Net energy (received energy – delivered energy)	IEC 62053-22 Class 0.5S
	Total energy (received energy + delivered energy)	IEC 62053-22 Class 0.5S
	Power Quality	
	Voltage/Current harmonics [4]	Up to the 50th order
	Voltage/Current THD [5], Current TDD [6]	Up to the 50th order
	Crest factor [4, 7], K-factor [4, 8]	-
	Voltage/Current unbalance [4]	IEC 61000-4-30, NEMA MG1
	Voltage/Current phasor [4]	-
	Voltage/Current waveforms [4]	64 samples/cycle, 2 cycles of waveforms
	Events	
	Detection of fuse fail, phase open, blackout, and over-temperature events	
	Maximum number of event logs	50
	Temperature	
	Temperature [9] of Accura 3300E	Measured with the temperature sensor integrated into the MCU

	Communication	
	RS-485	Performs communication with the host system using the Modbus RTU protocol
		1,200 / 2,400 / 4,800 / 9,600 / 19,200 / 38,400 / 57,600 / 115,200 [10] bps
	Power Supply	
	Power supply voltage [11]	AC 100 – 240 V 50/60 Hz, DC 100 – 300 V, CAT II

1. The residual voltage is the RMS voltage value calculated by summing three-phase voltages in the time domain.
2. The residual current is three times the zero-sequence current value of fundamental components. When harmonics are absent, the residual current equals the RMS current obtained by summing three-phase currents in the time domain
3. The predicted demand is calculated based on thermal demand, and the response speed of thermal demand can be set. For details, refer to「Chapter 4 Measurements > Demand Measurements」
4. The data are not displayed on the LCD screen of 3300E and are provided only via communication.
5. THD(Total Harmonic Distortion). Voltage THD: $\frac{\sqrt{\sum_{k=2}^{50} V_k^2}}{V_1}$, Current THD: $\frac{\sqrt{\sum_{k=2}^{50} I_k^2}}{I_1}$
6. TDD(Total Demand Distortion). Current TDD: $\frac{\sqrt{\sum_{k=2}^{50} I_k^2}}{I_L}$

where I_L can be set to the peak demand current(default) or the rated current(only via communication).

7. Crest factor: $\frac{I_{peak}}{I_{rms}}$, 8. K-factor: $\frac{\sum_{k=1}^{50}(k\, I_k)^2}{\sum_{k=1}^{50} I_k^2}$
9. It is not for fire monitoring but for reference only.
10. At 115,200 bps, a master can communicate with up to 16 slave devices, and the maximum distance between the master and the slave is 600 m.
11. The device conforms to UL standards for the tested AC power supply voltage.

Chapter 2 Installation

Environmental Conditions

Accura 3300E should be mounted in a dry, dirt free location away from heat sources and very high electric fields. To operate the product properly, environmental conditions as below should be considered.

Environmental Conditions	Description
Operating temperature	-20 – 70°C (-4 – 158 ℉)
Operating environment	Pollution degree 2
Safe operating temperature [1]	-20 – 60°C (-4 – 140 ℉)
Storage temperature	-40 – 85°C (-40 – 185 ℉)
Operating humidity	5 – 95 % (non-condensing)
Operating altitude	Up to 2,000 m
Ingress protection against water and dust	IEC 60529 IP54

1. Compliant with UL 61010-1(3rd edition)

Caution

- The meter should be installed in a switchgear cabinet or similar enclosure to prevent users from accessing the terminal strips on the meter and to minimize exposure to external environments after installation.
- Install the product in a dry location free from dirt, oil and corrosive vapors. Once the product is installed, there is no need to clean it.

Before Installation

Before installing the Accura 3300E, observe the following instructions and safety precautions.

Caution

You should power up the device after voltage and current wiring is completed.

Product Appearance

Fig 2.1 Accura 3300E Front

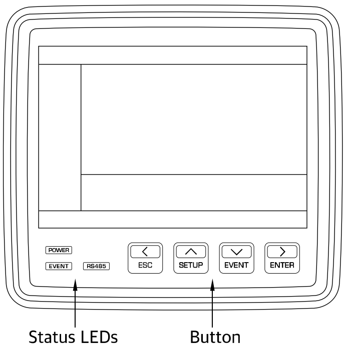


Fig 2.2 Accura 3300E Side

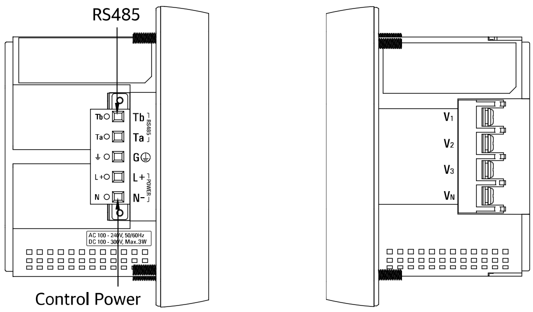
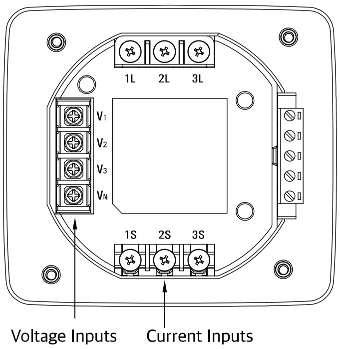
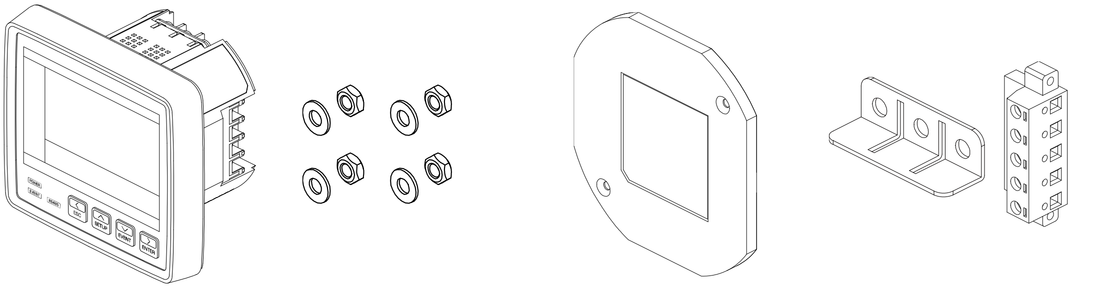


Fig 2.3 Accura 3300E Rear



Components

Fig 2.4 Accura 3300E Components



Accura 3300E | Four washers/four nuts | Rear protection cover | Short bar | Connector

Dimensions

Fig 2.5 Accura 3300E Front Dimensions

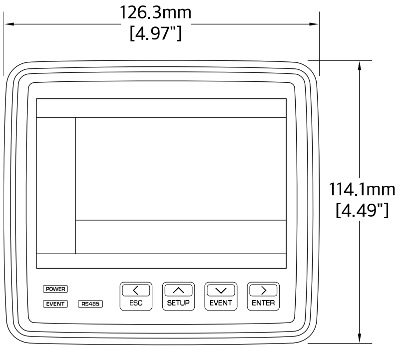


Fig 2.6 Accura 3300E Side Dimensions

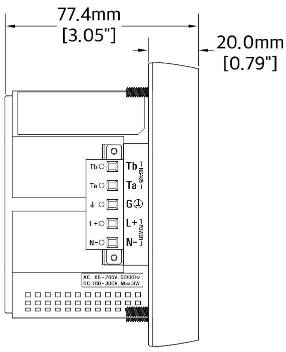
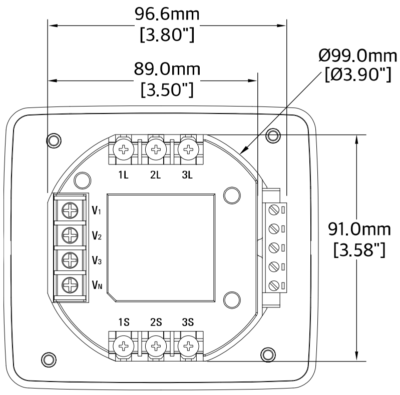


Fig 2.7 Accura 3300E Rear Dimensions



Product Dimensions

Device Type	Model	Dimensions [mm]
Digital Power Meter	Accura 3300E	126.3 W x 114.0 H x 77.4 D

Step 1: Installation

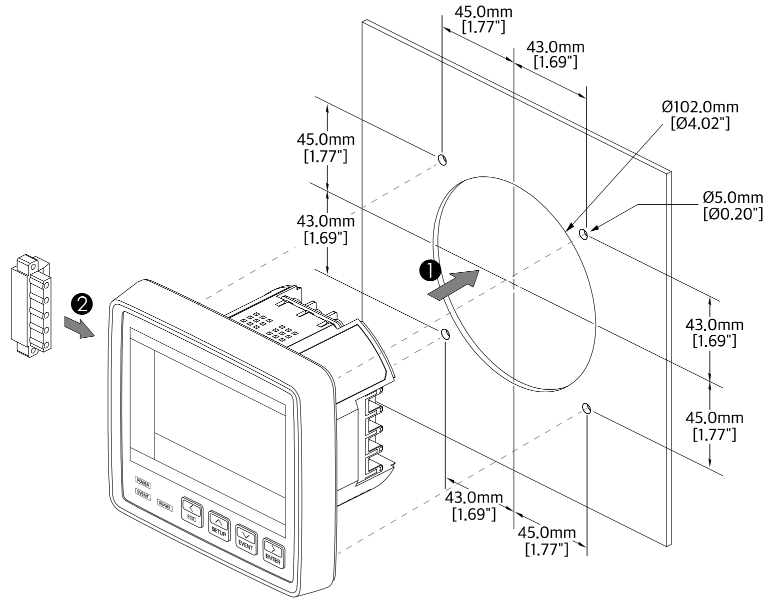
Mounting Accura 3300E on the Panel

Accura 3300E is mounted at the front of the switchgear panel.

① Insert the meter into the panel cutout.

② Attach the connector to the side of Accura 3300E and then secure the meter to the panel using mounting nuts and washers.

Fig 2.8 Mounting Accura 3300E on the Panel

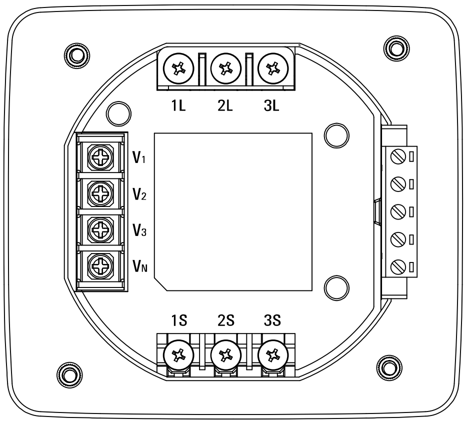


Note

The distance between each cutout(Ø5.0 mm) and the middle cutout(Ø102.0 mm) of the panel varies.

Step 2: Voltage/Current Wiring

Fig 2.9 Voltage/Current Wiring



Voltage Input

Item	Description
Port name	V_1, V_2, V_3, V_N
Connector type	Terminal block [1]
Wire specification	0.34 – 2.5 mm^2 (22 – 14 AWG), cooper
Wire temperature rating	Above 70 ˚C
Measurement category	CAT Ⅲ
Withstand voltage	AC 3,000 V RMS, 60 Hz for 1 minute
Impedance	10 MΩ/phase
Burden	0.01 VA/phase @ 220 V
Wiring method	3P4W, 3P3W, 1P3W, 1P2W
Rated voltage (Un)	AC 230 V L-N(Line-to-neutral)
Measuring range (accuracy guaranteed)	AC 35 – 277 V L-N, AC 60 – 480 V L-L(Line-to-line)
Minimum measured voltage	AC 5V L-N, AC 9 V L-L
Frequency range	42 – 65 Hz

1. The maximum screw tightening torque of the barrier-type terminal block is 1.1 Nm(12 kgf-cm, 10 in-Ib).

Note

If the voltage input gets out of the measurement range, it can affect accuracy. External PTs(Potential Transformers) should be used when measured voltages exceed the specified input voltage range.

Caution

When using external PTs, a fuse should be installed at the PT secondary.

Current Input

Item	Description	
Port name	1S, 2S, 3S (ports on the power supply side), 1L, 2L, 3L (ports on the load side)	
Connector type	Terminal block [1, 2]	
Wire specification	2.5 – 6.0 mm^2 (14 – 10 AWG), cooper	
Wire temperature rating	Above 70 ˚C	
Rating	5 A nominal / 10 A full scale 3~	Accura 3300E-5A
	1 A nominal / 2 A full scale 3~	Accura 3300E-1A
Measuring range (Accuracy guaranteed)	1 %In – 200 %In [3]	Accura 3300E-5A
	5 %In – 200 %In [3]	Accura 3300E-1A
Minimum measured current	5 mA	Accura 3300E-5A
		Accura 3300E-1A
Burden	Up to 0.01 VA/phase @10 A	

1. The maximum screw tightening torque of the barrier-type terminal block is 1.4 Nm(14 kgf-cm, 12 Ibf-in).
2. When connecting wires to the terminal block, use UL listed lugs.
3. The rated current In is 5 A/1 A.

Wiring Considerations

Caution

It is recommended you should use CTT terminal blocks in order to prevent accidents caused by CTs(Current Transformers) and identify device failures in advance. If current is flowing in the primary winding of a CT, opening the CT secondary can cause serious accidents resulting from high voltages. Make sure to use UL2808 listed CTs(reinforced insulation) approved by US standards.

The installation and operation of the product should be performed only by qualified, competent personnel that have received appropriate training for high voltage and current equipment.

Caution

Failure to observe the following instructions may result in severe injury or death.

- During normal operation of the product, hazardous voltages are present on its terminal strips that connect the PT, CT, voltage input, power, and external I/O circuits. The secondary of the PT/CT may generate fatal voltages/currents resulting from the primary of the PT/CT.
- Standard safety precautions should be followed while installing or maintaining the product, including removing PT fuses and shorting CT secondaries.
- Install the product in a panelboard to prevent users from accessing terminal strips after completing wiring of the product.

Caution

Observe the following instructions, or permanent damage may occur to the product.

- Do not apply voltages and currents that exceed the input ratings of the PT and the CT to the product.
- Using the device for purposes not specified by the manufacturer can cause severe damage to the product.
- Connect the ground terminal to the earth ground to protect the device from noise and surge. Failure to properly connect the chassis ground will void the warranty of the product.

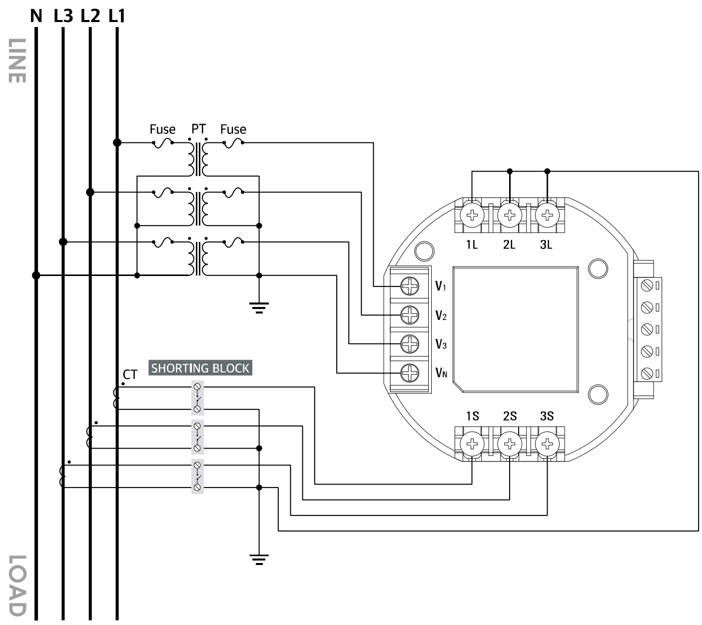
Wiring Diagram Using External PTs

Caution

Shorting blocks should be used in case of wiring the meter and the CT secondary.

- When there is the primary current of the CT, opening the CT secondary can cause serious accidents resulting from high voltages.
- When wiring devices without a CT, you should follow the guidelines provided by Rootech.

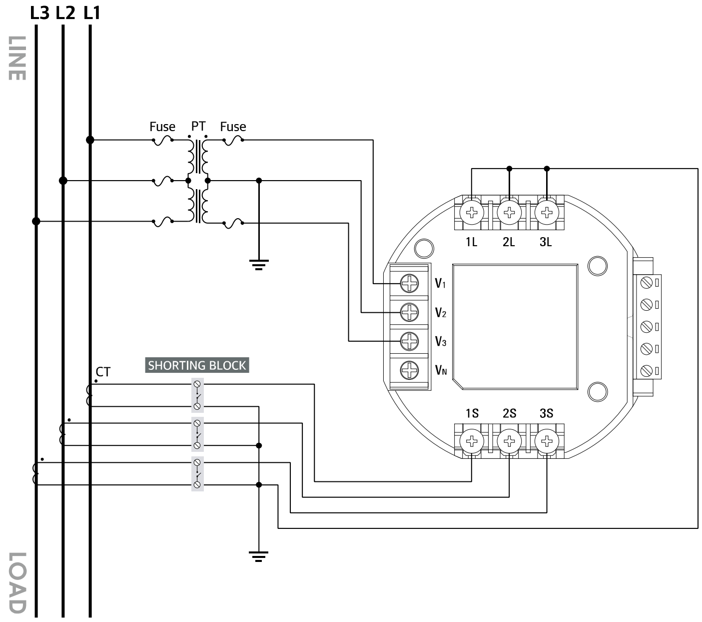
Fig 2.10 3P4W Connection with 3 PTs and 3 CTs, Wiring Mode(Conn)= 3P4U



Note

Select "3P4U" from the "Wiring Mode" setup menus.

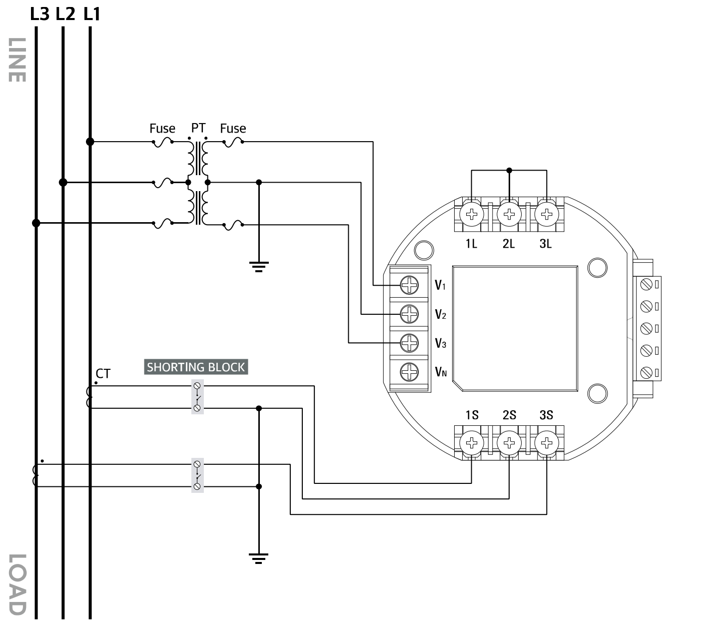
Fig 2.11 3P3W Connection with 2 PTs and 3 CTs, Wiring Mode(Conn) = 3P3U



Note

Select "3P3U" from the "Wiring Mode" setup menus.

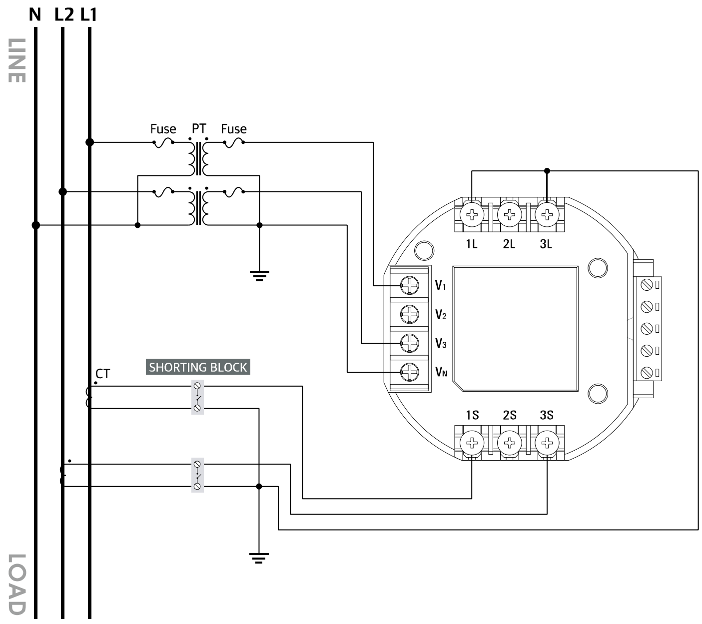
Fig 2.12 3P3W Connection with 2 PTs and 2 CTs, Wiring Mode(Conn) = 3P3U



Note

Select "3P3U" from the "Wiring Mode" setup menus.

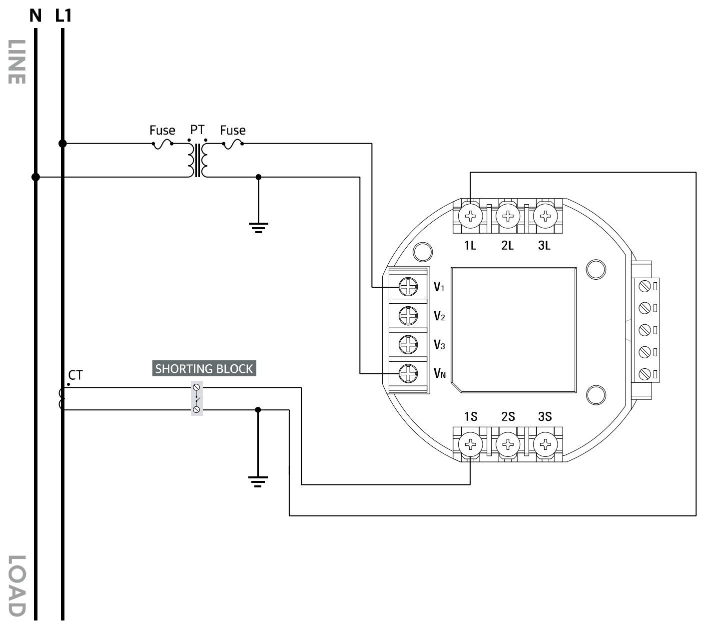
Fig 2.13 1P3W Connection with 2 PTs and 2 CTs, Wiring Mode(Conn) = 1P3U



Note

Select "1P3U" from the "Wiring Mode" setup menus.

Fig 2.14 1P2W Connection with 1 PT and 1 CT, Wiring Mode(Conn) = 1P2U

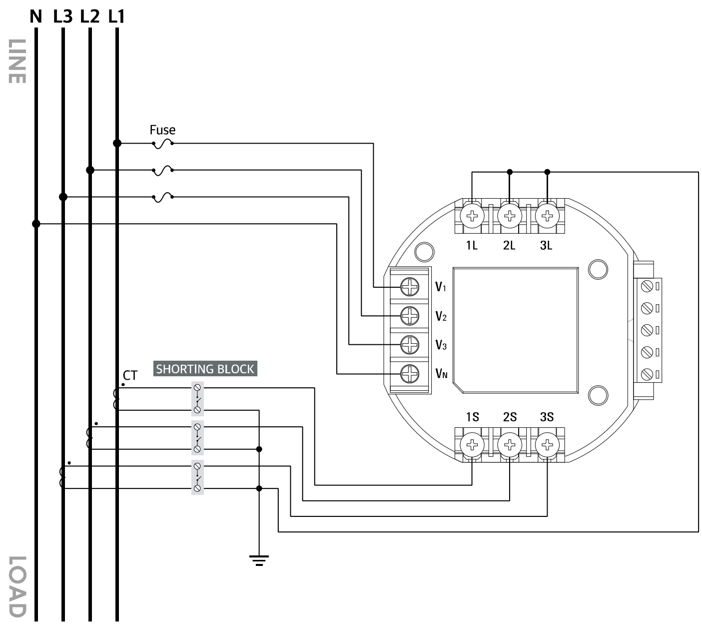


Note

V_1 and V_N terminals should be used. Select "1P2U" from the "Wiring Mode" setup menus.

Wiring Diagram Without Using External PTs

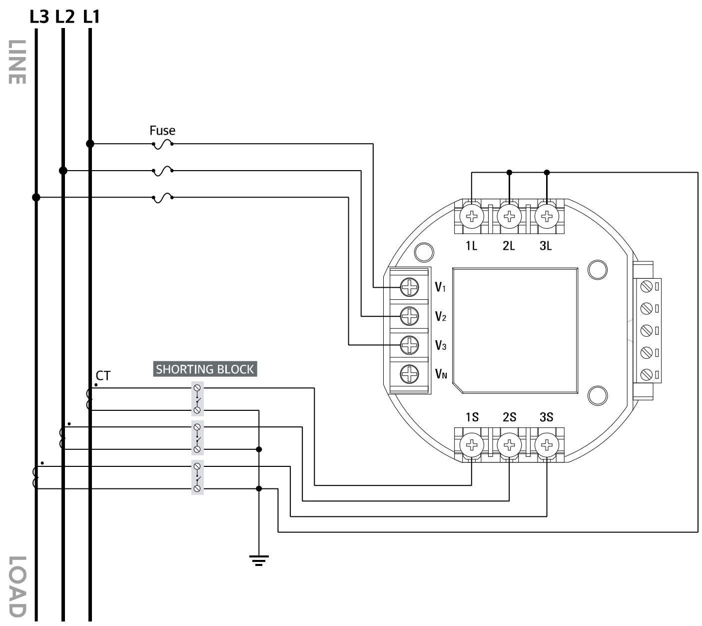
Fig 2.15 3P4W Direct Connection with 3 CTs, Wiring Mode(Conn) = 3P4U



Note

Select "3P4U" from the "Wiring Mode" setup menus.

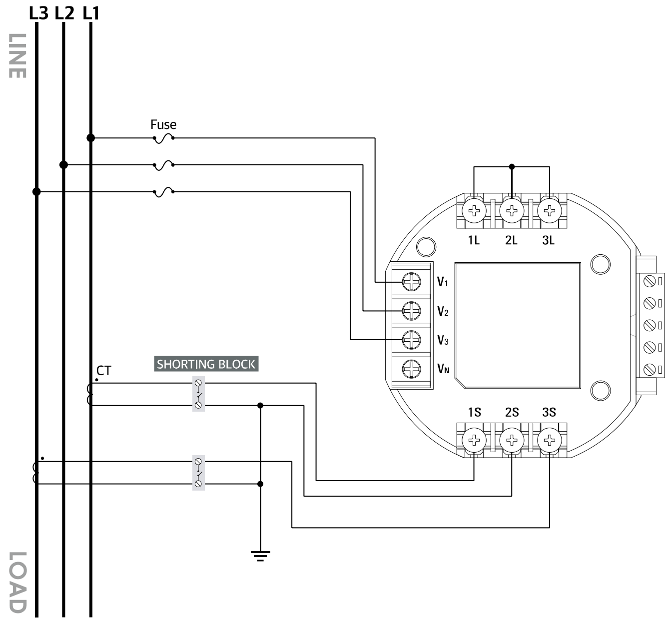
Fig 2.16 3P3W Direct Connection with 3 CTs, Wiring Mode(Conn) = 3P3U



Note

Select "3P3U" from the "Wiring Mode" setup menus.

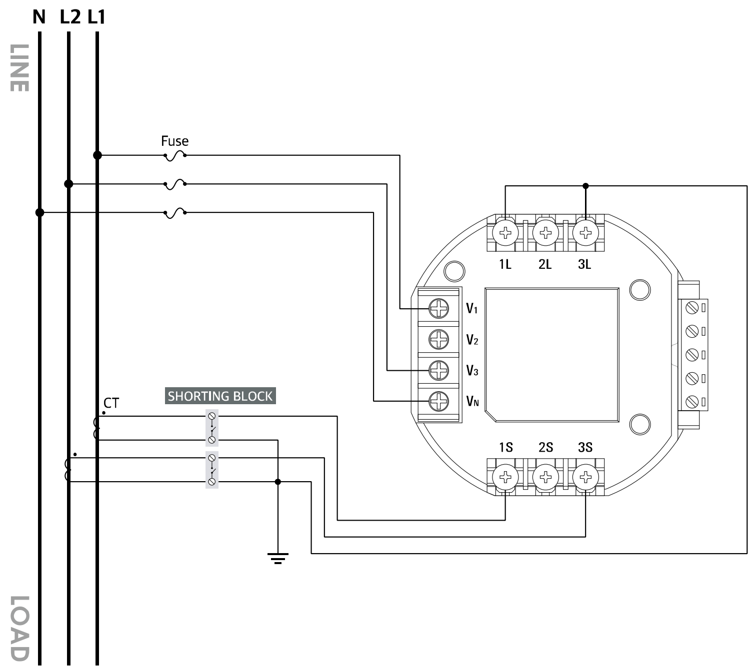
Fig 2.17 3P3W Direct Connection with 2 CTs, Wiring Mode(Conn) = 3P3U



Note

Select "3P3U" from the "Wiring Mode" setup menus.

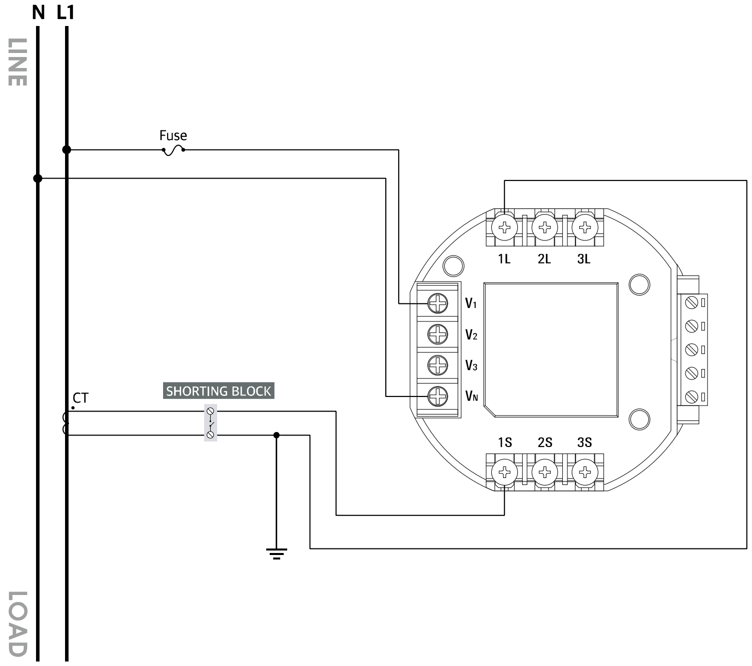
Fig 2.18 1P3W Direct Connection with 2 CTs, Wiring Mode(Conn) = 1P3U



Note

Select "1P3U" from the "Wiring Mode" setup menus.

Fig 2.19 1P2W Direct Connection with 1 CT, Wiring Mode(Conn) = 1P2U



Note

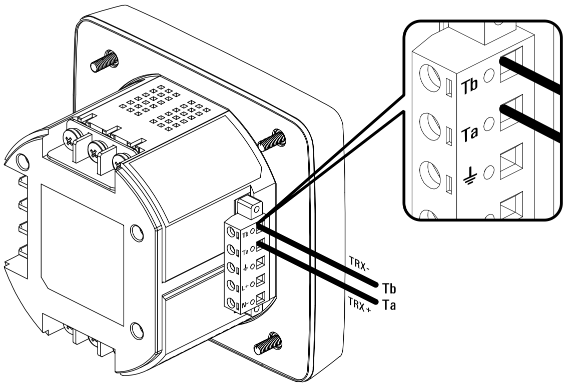
V_1, and V_N terminals should be used. Select "1P2U" from the "Wiring Mode" setup menus.

Step 3: Wiring Accura 3300E for External Communication

RS-485 Communication

Accura 3300E supports one RS-485 communication port.

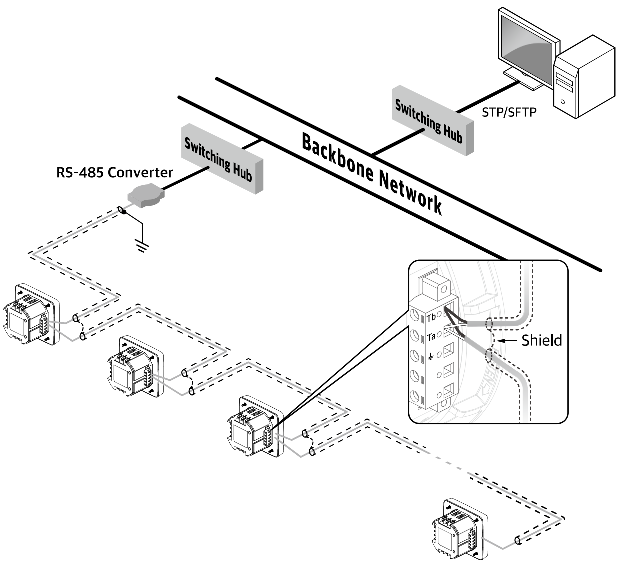
Fig 2.20 RS-485 Communication of Accura 3300E



Item	Description
Port name	Ta, Tb
Connector type	Screw-type terminal (pluggable)
Protocol	Modbus RTU
Wire specifications	UL 2919 RS-485 1P/2P 24 AWG
Maximum cable length	1,219 m (4,000 ft)
Bit rate	1,200 / 2,400 / 4,800 / 9,600 / 19,200 / 38,400 / 57,600 / 115,200 bps [1]
Maximum number of connectable devices	32 per bus

1. At 115,200 bps, a master can communicate with up to 16 slave devices, and the maximum distance between the master and the slave is 600 m.

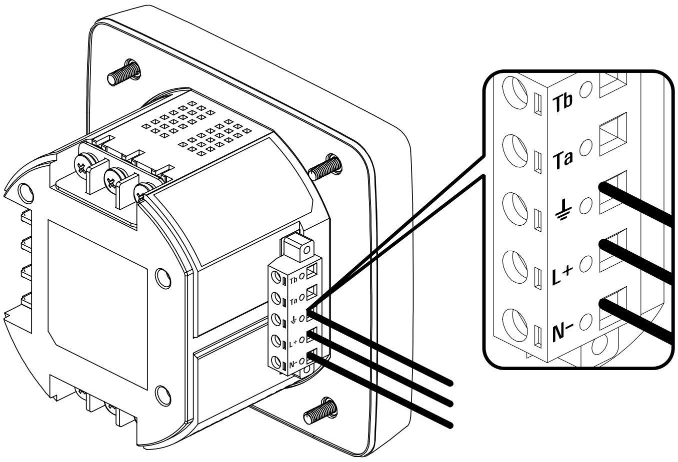
Fig 2.21 Connecting Accura 3300E Devices for RS-485 Communication



* To minimize noise caused by ground loops, a shielded cable should be grounded at only one end as shown in the figure.

Step 4: Power Supply and Ground Wiring

Fig 2.22 Power Supply Wiring of Accura 3300E



Power Supply Wiring

Item	Description	
Port name	L+, N-	
Connector type	Screw-type terminal (pluggable) [1]	
Wire specification	0.25 – 4.0 mm² (24 – 12 AWG), cooper	
Wire temperature rating	Above 70 ˚C	
Power supply voltage (Us) [2]	AC 100 – 240 V 50/60 Hz, DC 100 – 300 V, CAT II	
Withstand voltage	AC 3,000 V RMS, 60 Hz for 1 minute	
Operating environment	Pollution degree 2	
Power consumption	During normal operation	Maximum 3 W
	When the device is powered on	Maximum inrush current: peak 20 A, duration: within 1 msec [3]

1. The maximum screw tightening torque of the screw-type terminal block is 0.51 Nm(5.2 kgf-cm, 4.5 Ibf-in).
2. The device conforms to UL standards for the tested AC power supply voltage.
3. When installing fuses on the power line, it is recommended to use fuses that are UL-listed with an RK5 class or higher and are rated at 2 A/500 V.

Ground Wiring

Item	Description
Port name	⏚
Connector type	Screw-type terminal (pluggable)
Wire specification	0.25 – 4.0 mm² (24 – 12 AWG)

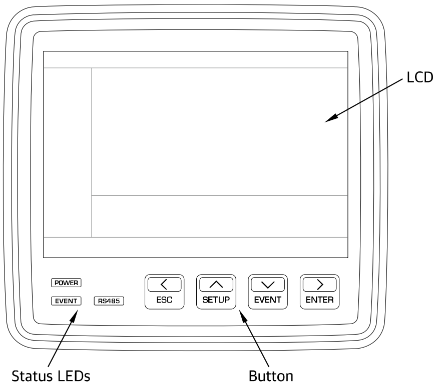
Note

Wire the meter's ground terminal ⏚ to the earth ground of the distribution panel.

Chapter 3 Operation/Setup

The front panel of Accura 3300E consists of the 4.7" LCD screen which shows measurement and setup data, three LEDs which display power supply and communication status, and four buttons used to select all operation modes.

Fig 3.1 Front Panel of Accura 3300E



LEDs

The three LEDs located at the bottom left corner of the front panel display the operation state of Accura 3300E.

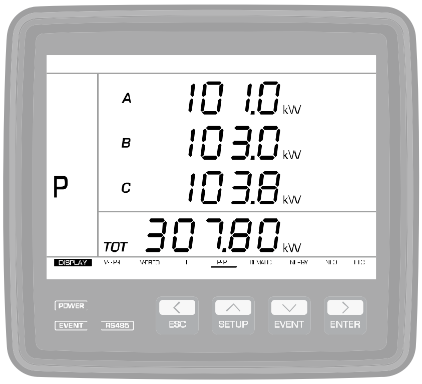
LED Name	Color	Description
POWER	Green	Turned on when the power supply of Accura 3300E operates normally
EVENT	Red	Blinks when an event occurs
RS485	Yellow	Blinks during RS-485 communication

Button Operations

There are four buttons on the front panel: **Left[<]/ESC, Up[^]/SETUP, Down[v]/EVENT**, and **Right[>]/ENTER** buttons. When you short press the buttons, they operate as directional buttons. When you press the buttons for more than 1 second, the device performs the specialized functions of the **ESC, SETUP, EVENT**, and **ENTER** buttons.

In the display mode which displays measurement data, you can navigate to all measurement display screens using the directional buttons, the **Left[<], Up[^], Down[v]**, and **Right[>]** arrow buttons. At this time, the displayed menu appears in the navigation bar at the lower part of the LCD screen, enabling you to easily check the location of the currently displayed screen.

Fig 3.2 Screen in Display Mode

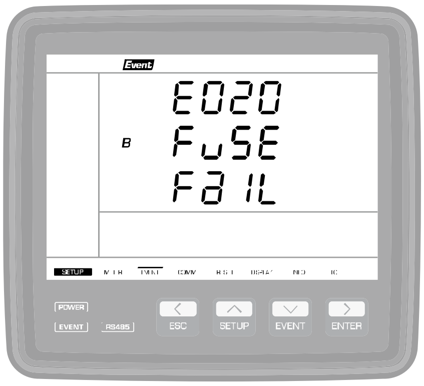


By long pressing the **SETUP** button, you can enter the setup mode and configure the device's wiring mode, PT rating, RS-485 communication, events, etc. In the setup mode, you can check and change setup data using the **Left[<], Up[^], Down[v], Right[>], ESC,** and **ENTER** buttons. At this time, the SETUP menu is displayed in the navigation menu bar at the lower part of the LCD screen, enabling you to easily check the location of the currently displayed screen.

Fig 3.3 Screen in Setup Mode

When an event such as a fuse fail, phase open, blackout, and an over-temperature event occurs, the EVENT LED and LCD backlight blink to let users easily recognize the occurrence of the event. You can enter the event log mode by long pressing the **EVENT** button. In the event mode, you can clear the alarm on the event that has occurred and check the event log. Even when an event does not occur, you can check the event logs by long pressing the **EVENT** button. In the event mode, the "Event" segment in the top menu bar lights up.

Fig 3.4 Screen in Event Mode

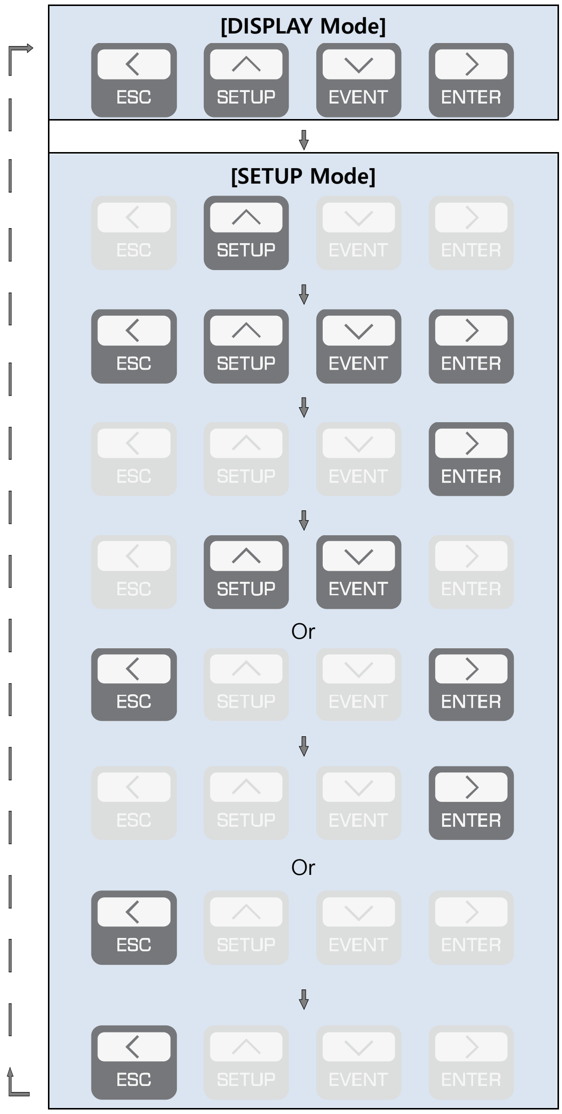


Mode	Description
DISPLAY mode	Displays measurement data As the displayed menu appears in the navigation menu bar at the lower part of the LCD screen, you can easily check the location of the currently displayed screen.
SETUP mode	Enables users to check and change setup items(wiring mode, PT rating, RS-485 communication, events, etc.) As the displayed menu appears in the navigation menu bar at the lower part of the LCD screen, you can easily check the location of the currently displayed screen.
EVENT mode	Displays the event logs. In the event mode, the “Event” segment in the top menu bar lights up to indicate the event mode.

Note

In the setup mode, the navigation menu appearing at the lower part of the LCD screen changes from “DISPLAY” to “SETUP”. When an event occurs, the backlight of the LCD screen blinks. The LCD backlight timeout is 5 seconds by default, and users can set it to a value up to 9,999 minutes.

The following table displays the functions performed by short pressing and long pressing the buttons.

Button		Function
	Press [1]	Select the display screen with the up/ down/left/right arrow button.
	Long Press [2]	Move to the setup mode.
	Press	Select the setup menu.
	Long Press	Make the displayed setup value changeable. (The screen blinks.)
	Press	Select the value to set up.
	Press	Move to the next digit to set up.
	Long Press	Save the set value. (The screen stops blinking.)
	Long Press	Do not save the set value. (The screen stops blinking.)
	Long Press	Exit the setup mode and move to the display mode.

1. The "Beep" sound occurs when the button is pressed.
2. Press the button for more than one second. The "Beep" sound occurs in one second.

Note

If there is no button operation for 10 minutes [1] in the setup mode, the screen automatically returns to the display mode. If there is no button operation for 1 minute in the edit mode, the screen automatically returns to the setup mode.

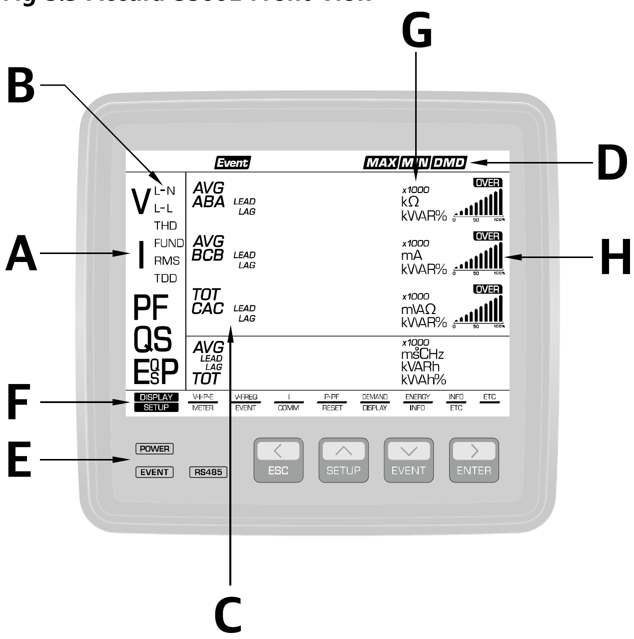
1. It can be set via communication, and the default value is 10 minutes.

DISPLAY Mode

Front View

Displayed segments on the screen may differ according to the hardware version of Accura 3300E. However, displayed parameters are the same.

Fig 3.5 Accura 3300E Front View



D : Top Menu Bar		
Event	Event	Displays the event mode
MAX	Maximum	
MIN	Minimum	
DMD	Demand	

E : LEDs at the Bottom Left Corner of the Front Panel	
POWER	Indicates the device is powered on
EVENT	Blinks when an event occurs
RS485	Blinks during RS-485 communication

A	
V	Voltage
I	Current
P	Active Power
Q	Reactive Power
S	Apparent Power
E	Active Energy
PF	Power Factor

B	
L-N	Line-to-Neutral
L-L	Line-to-Line
THD	Total Harmonic Distortion [%]
FUND	Fundamental
RMS	Root Mean Square
TDD	Total Demand Distortion [%]

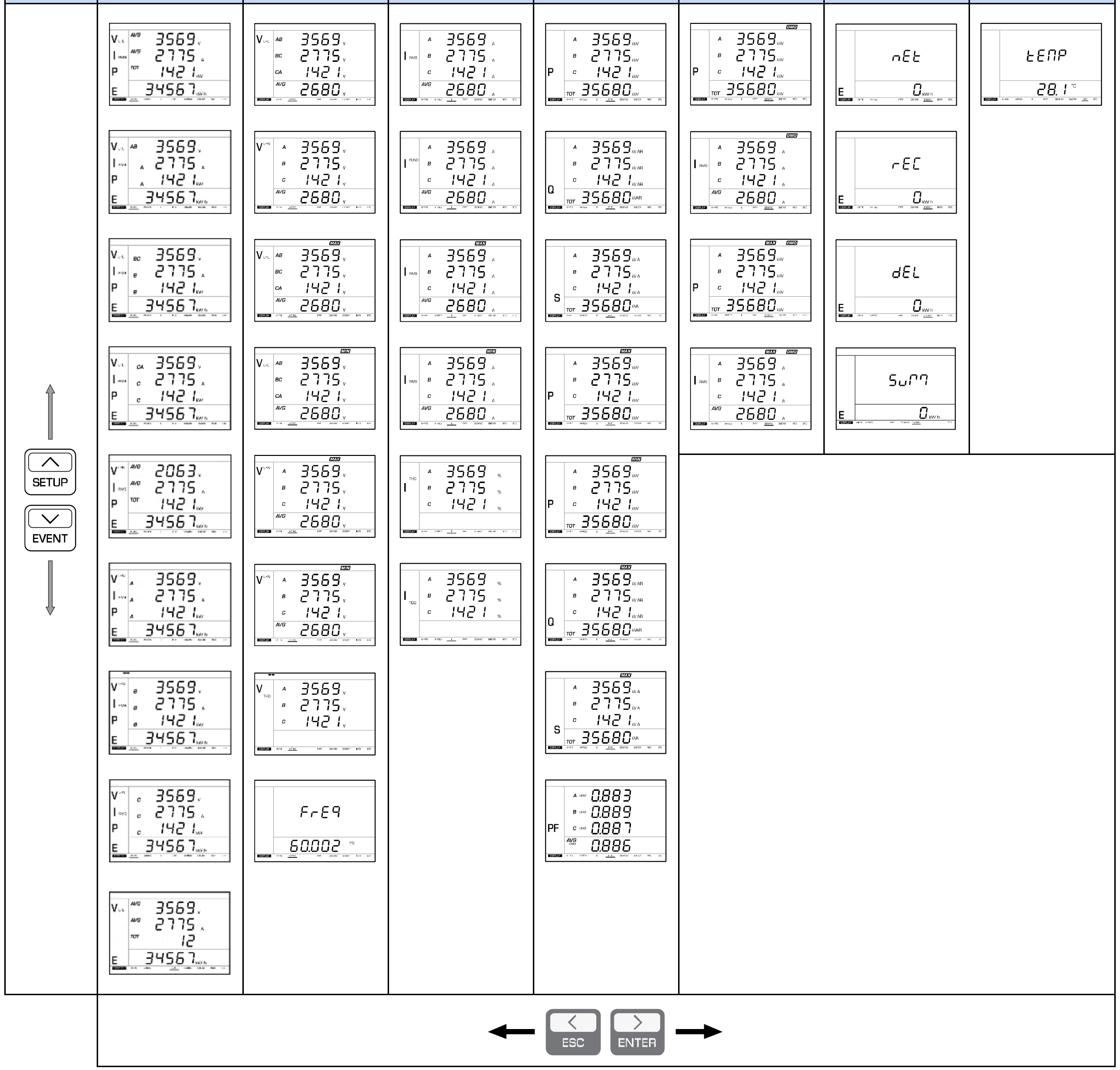
C	
A	Phase A
B	Phase B
C	Phase C
AB	Line-to-Line AB
BC	Line-to-Line BC
CA	Line-to-Line CA
LEAD	Lead
LAG	Lag
AVG	Average of three-phase values
TOT	Sum of three-phase values

F: Navigation Menu	
DISPLAY	
V·I·P·E	Voltage, Current, Power, Energy
V·FREQ	Voltage, Frequency
I	Current
P·PF	Power, Power Factor
DEMAND	Demand Power, Demand Current
ENERGY	Energy (Net, Received, Delivered, Total)
ETC	Temperature of the device
SETUP	
METER	Setup for measurement parameters
EVENT	Setup for events
COMM	Setup for RS-485 communication
RESET	Setup for the reset function
DISPLAY	Setup for the LCD display
ETC	Meter information such as serial number, version, etc.
G: Unit	
H: Graph	

Display Screen

Accura 3300E displays information on measurement parameters.

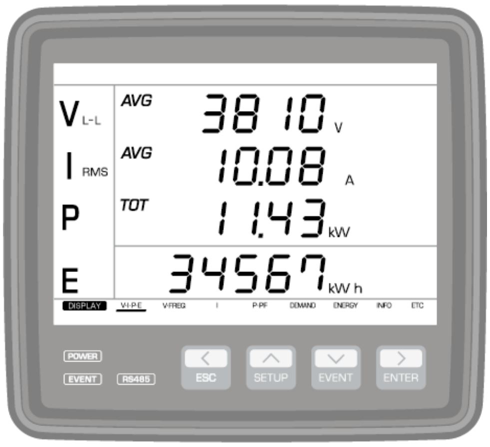
Column	Voltage/Current/ Power/Energy	Voltage/ Frequency [2]	Current	Power/ Power Factor	Demand	Active Energy	Temperature
	V·I·P·E	V·FREQ	I	P·PF	DEMAND	ENERGY	ETC



Measurement Parameter Display Screen

The screen shown first in the Display mode displays V·I·P·E: the average values of three-phase voltage and current, the total three-phase active power, and energy. The screen displays the lower 5 digits of the total energy value. Refer to the「DISPLAY > Enrg dISP」 screen in the Setup mode for details on how the energy value is displayed on the first screen.

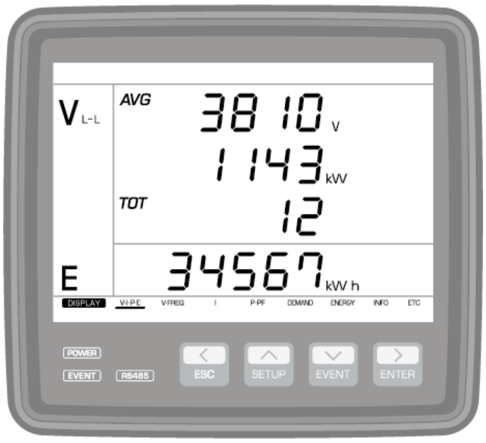
Fig 3.6 First Screen Appearing in the First Column of Display Mode



By pressing the **Down[v]** arrow button on the first screen, you can see the line-to-line voltage value between the phase A and B, and the current, power and energy values of phase A. By pressing the **Down[v]** arrow button again, you can see measurement values of other phases. You can also check the average value of line-to-neutral voltage and measured values for each phase. By pressing the **Up(^)** arrow button, you can see measurement values by moving between screens in the reverse direction.

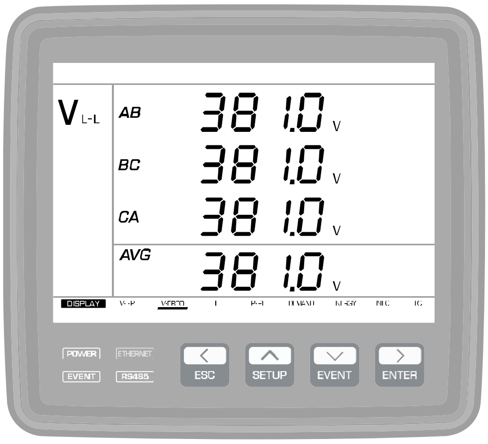
The last screen appearing in the first column displays the average voltage value of three phases, the total power, and energy(V·P·E). This screen displays the total energy value with nine digits appearing in the third and the fourth lines.

Fig 3.7 Last Screen Appearing in the First Column of DISPLAY Mode



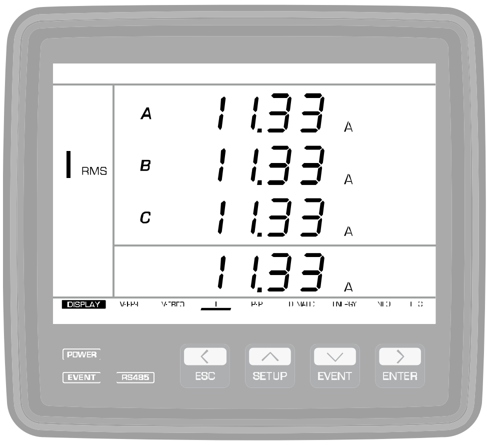
By pressing the **Right[>]** arrow button on the first display screen, you can see detailed voltage data. The first screen appearing in the voltage column displays each line to line voltage value and average value. By pressing the **Down[v]** arrow button, you can check line-to-neutral voltage, the maximum and minimum line-to-line voltage values, the maximum and minimum line-to-neutral voltage values, voltage THD, and frequency in sequence. If you press the **Up[v]** arrow button, you can see measured values in reverse order.

Fig 3.8 Line-to-Line Voltage Display Screen



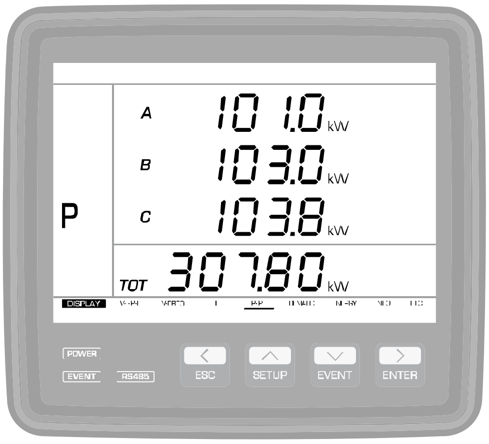
By pressing the **Right[>]** arrow button again on the voltage display screen, you can see detailed current data. The first screen appearing in the current column displays the true RMS value of the phase current per phase and the average value. By pressing the **Down[v]** arrow button, you can check the phase current of the fundamental component and the average value, the maximum and minimum phase current values, current THD, current TDD, residual current, zero-sequence/negative-sequence unbalance, and unbalance according to the NEMA standard in sequence. If you press the **Up[v]** arrow button, you can see measured values in reverse order.

Fig 3.9 Phase Current Display Screen



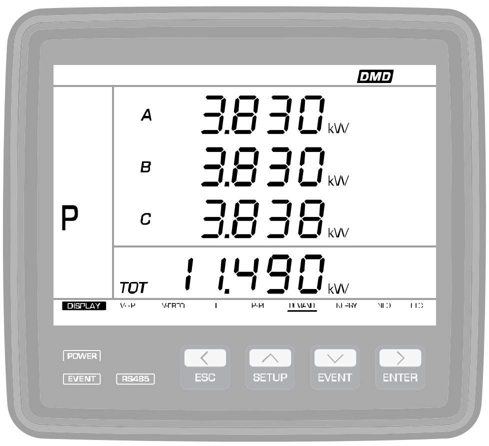
By pressing the **Right[＞]** arrow button again on the current display screen, you can see detailed data on power and power factor. The first screen appearing in the power column displays the active power value per phase and the total active power value. By pressing the **Down[∨]** arrow button, you can check reactive power[Q], apparent power[S], the maximum and minimum active power, the maximum reactive power, the maximum apparent power and power factor(PF) values in sequence. If you press the **Up[∨]** arrow button, you can see measured values in reverse order.

Fig 3.10 Power Display Screen



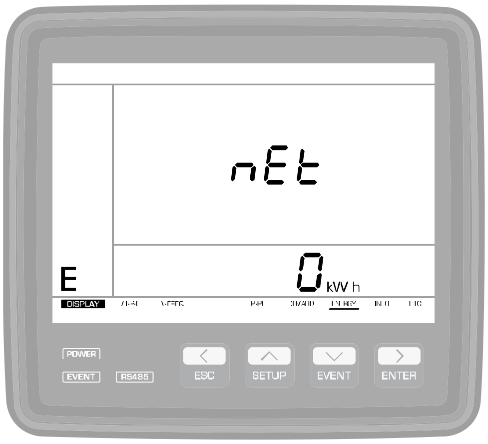
By pressing the **Right[＞]** arrow button again on the power display screen, you can see detailed data on demand power and demand current. The first screen appearing in the demand column displays the demand active power per phase and the total demand active power value which is the sum of three-phase demand active power. By pressing the **Down[∨**] arrow button, you can check demand current, the maximum demand active power and demand current values in sequence. If you press the **Up[∨]** arrow button, you can see measured values in reverse order.

Fig 3.11 Demand Active Power Screen



By pressing the **Right[>]** arrow button again on the demand display screen, you can see detailed data on energy. The first screen appearing in the energy column displays the net energy. By pressing the **Down[v]** arrow button, you can check the received energy, delivered energy and sum of energy in sequence. If you press the **Up[v]** arrow button, you can see measured values in reverse order.
Net energy is the value obtained by subtracting the delivered energy from the received energy, and sum of energy is the value obtained by adding up the received energy and delivered energy.

Fig 3.12 Net Energy Display Screen on the Energy Display Screen



By pressing the **Right[>]** arrow button again on the energy display screen, you can navigate to the ETC screen. The ETC screen displays the temperature of Accura 3300E measured by the temperature sensor integrated into the MCU.

Fig 3.13 Temperature Display Screen on the ETC Display Screen

Event Mode

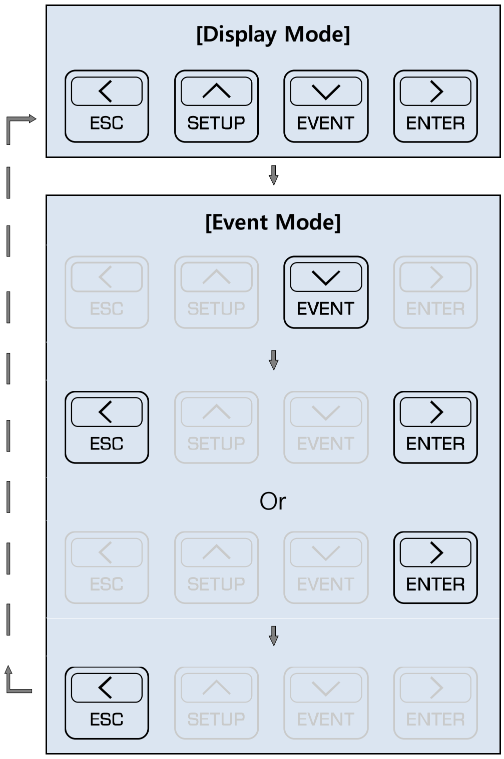
Generation of Event Alarms

Accura 3300E provides event alarms when the device detects events such as fuse fail, phase open, blackout, and over-temperature events. When an alarm is generated, the LCD backlight and the EVENT LED at the bottom left side of the device blink.

Clearing Event Alarms

When an event alarm is generated, the event alarm can be automatically cleared by long-pressing the **EVENT** button and entering the event mode. In the event mode, the "Event" segment in the top menu bar lights up and displays up to 50 event logs.

Button Operation



Button	Function
Press [1]	Select the display screen with the up/down/left/right arrow buttons.
Long Press [2]	Move to the event mode. (The latest event log is displayed.)
Press	Select an event log.
Long Press	Move to the latest event log.
Long Press	Exit the event mode and move to the display mode.

1. The "Beep" sound occurs when you press the button.
2. Press the button for more than one second. The "Beep" sound occurs in one second.

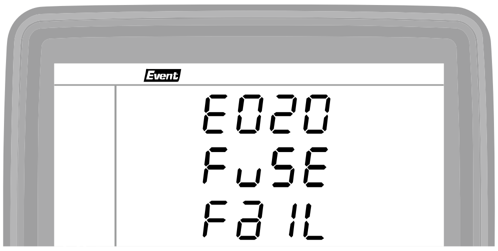
Note

When an event occurs, the LCD backlight timeout is 5 seconds by default, and users can set the timeout to a value up to 9,999 minutes. The screen automatically returns to the display mode if there is no button operation for 10 minutes in the event mode.

Event Log Number

Event log numbers are given in sequence with numbers from 'E000' to 'E999' according to the order in which events occur. When the log number reaches 'E999', 'E000' is given again as the 1st log number. You can save up to 50 event logs.

Fig 3.14 Example of Event Log Number



The following table shows that one additional event log is generated in the saved event logs from 'E051' to 'E100'.

Event Log [E051 - E100]		
E000	Start event log number	
E001		
⋮		
E050		
E051	Event log number N − 49 [1]	↑
E052	Event log number N − 48	50
⋮		
E100	Event log number N [1]	↓
E101		
⋮		
E998		
E999	Last event log number	

→

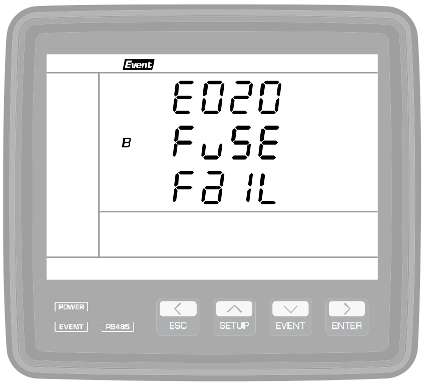
Event Log [E052 - E101]		
E000	Start event log number	
E001		
⋮		
E050		
E051		
E052	Event log number N − 49	↑
E053	Event log number N − 48	50
⋮		
E101	Event log number N	↓
E102		
⋮		
E999	Last event log number	

1. Event log numbers are as follows.

Event Log number N – 49	Oldest saved event
Event Log number N	Latest saved event

Fuse Fail Event Log

It displays the information when a fuse fail event occurs.



Display	
E[Event]	Event number
[d][d][d] [1]	000 – 999
B FuSE (Fuse)	The phase where the event occurs
FaIL(Fail)	(B phase) [2]
	Fuse fail

1. d → decimal number
2. The phase(A, B, C) where the event occurs is displayed.

Phase Open Event Log

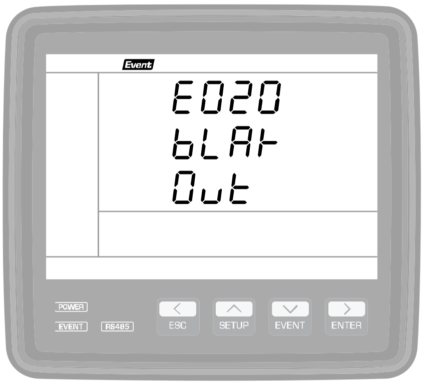
It displays the information when a phase open event occurs.

Display	
E[Event]	Event number
[d][d][d] [1]	000 – 999
B PHSE (Phase)	The phase where the event occurs
OPEn (Open)	(B phase [2])
	Phase open

1. d → decimal number
2. The phase(A, B, C) where the event occurs is displayed.

Blackout Event Log

It displays the information when a blackout event occurs.



Display	
E[Event]	Event number
[d][d][d] [1]	000 – 999
bLAk (Black)	A blackout event occurs.
Out (Out)	

1. d → decimal number

Over-Temperature Event Log

It displays the information when an over-temperature event occurs.

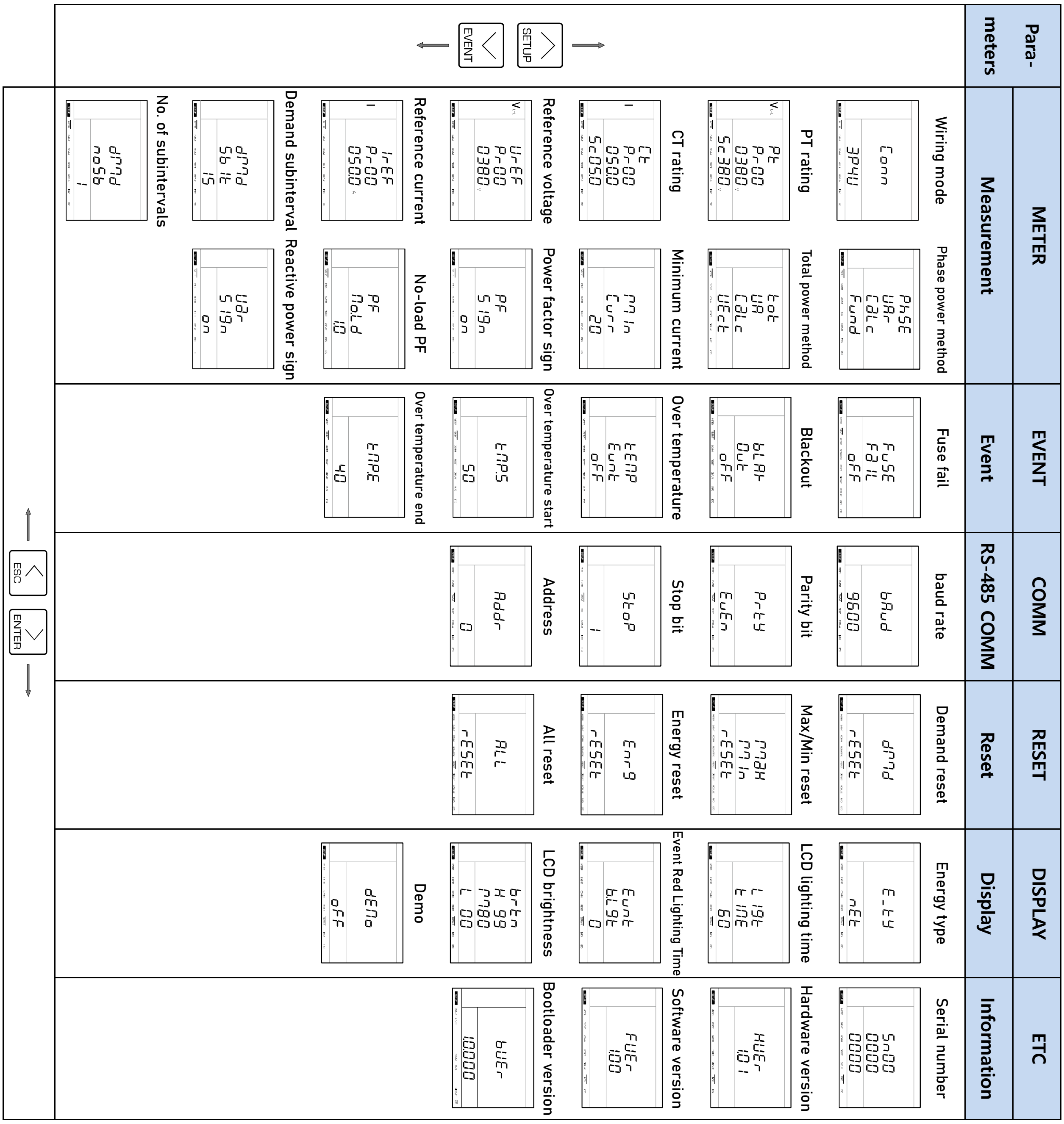
Display	
E[Event]	Event number
[d][d][d] [1]	000 – 999
tEnP (Temperature)	An over-temperature event occurs.
OvEr (Over)	

1. d → decimal number

SETUP Mode

To set the wiring mode, PT rating, RS-485 communication, and events of Accura 3300E and check setup data, you should enter the setup mode by long pressing the **SETUP** button. In the setup mode, you can check and change setup data with the **Left[<]**/**Up[^]**/**Down[v]**/**Right[>]** arrow, **ESC**, and **ENTER** buttons. The displayed menu appears in the navigation bar at the lower part of the LCD screen, enabling you to easily check the location of the currently displayed screen. The names of setup menus may differ according to the hardware version of Accura 3300E. However, displayed parameters are the same.

SETUP Menu



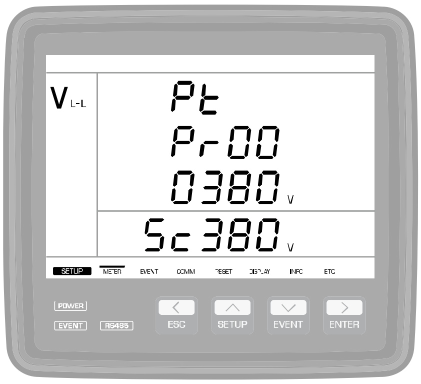
METER Setup

Connection Mode(Conn → Connection)

Setup Range
3P4U (3 Phase 4 Wire, default)
3P3U (3 Phase 3 Wire)
1P3U (Single Phase 3 Wire)
1P2U (Single Phase 2 Wire)

PT Rating(Pt → PT)

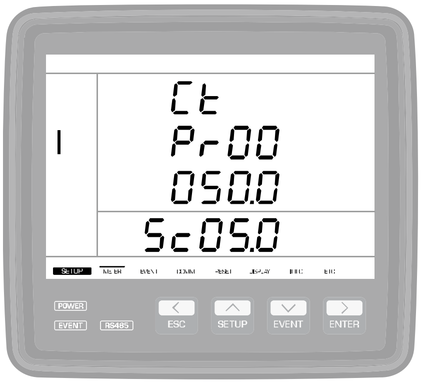
The ratio of the primary to secondary line-to-line voltage is used as the conversion ratio for the primary voltage. Primary and secondary line-to-line voltage values for each connection mode are as follows. In case of direct connections without using PTs, set the primary and secondary voltage values of the PT to the same values.



Setup Range	
Pr[Primary]	Primary voltage(line-to-line) rating of the PT
[d][d][d][d][d][d] [1]	000001 – 999999
[default 380]	
Sc[Secondary]	Secondary voltage(line-to-line) rating of the PT
[d][d][d]	001 – 999
[default 380]	

1. d → decimal number

CT Rating(Ct→ CT)

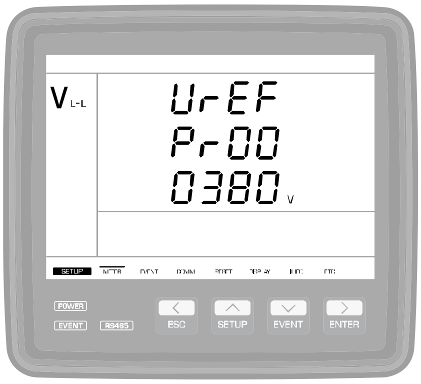


Setup Range	
Pr[Primary]	Primary current rating of the CT
[d][d][d][d][d].[d] [1]	00000.1 – 99999.9
[default 50.0]	Accura 3300E-5A
[default 10.0]	Accura 3300E-1A
Sc[Secondary]	Secondary current rating of the CT
[d][d].[d]	00.1 – 99.9
[default 5.0]	Accura 3300E-5A
[default 1.0]	Accura 3300E-1A

1. d → decimal number

Reference Voltage(UrEF)

The reference voltage is used as the reference value for bar graphs displayed on the screen. The line-to-line voltage value on the primary side **of the PT** should be entered as the reference voltage all the time regardless of voltage wiring modes.

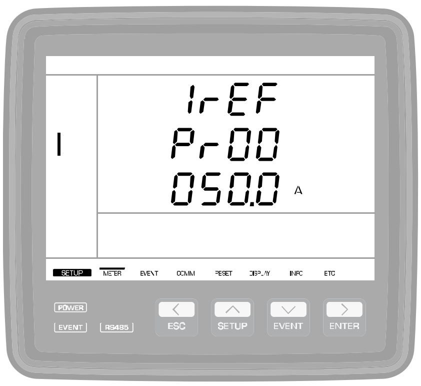


Setup Range	
Pr[Primary]	Reference line-to-line voltage value on the primary side of the PT
[d][d][d][d][d][d] [1]	000001 – 999999
[default 380]	

1. d → decimal number

Reference Current(IrEF)

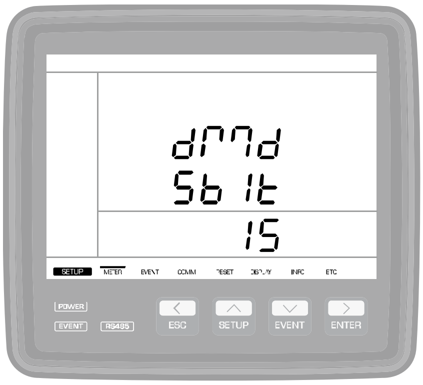
The reference current is used as the reference value for bar graphs displayed on the screen and TDD configurations.



Setup Range	
Pr[Primary]	Reference current on the primary side of the CT
[d][d][d][d][d].[d] [1]	00000.1 – 99999.9
[default 50.0]	Accura 3300E-5A
[default 10.0]	Accura 3300E-1A

1. d → decimal number

Demand Subinterval (dMd → Demand, SbIt → Subinterval)



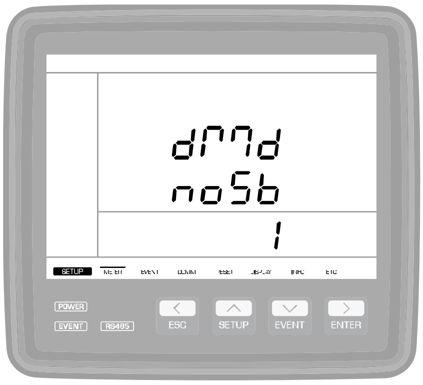
Setup Range
1 – 60 minute [default 15]

Number of Demand Subintervals (dMd → Demand, noSb → Number of Demand Subintervals)

Accura 3300E uses the "sliding window" method to calculate demand values.

Demand time[minute] = demand subinterval x number of demand subintervals

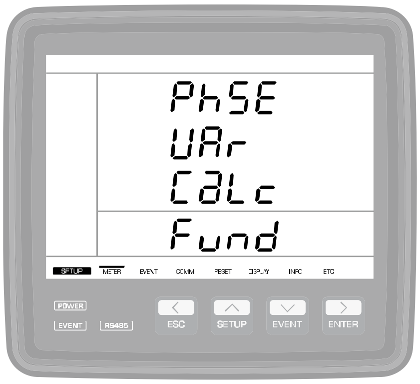
E.g.) In Korea(15-minute demand period), the demand subinterval is 15 minutes, and the number of demand subintervals is 1. Refer to Chapter 4 for details.



Setup Range
1 – 12 [default 1]

Calculation of Power Per Phase (PhSE Var CaLc → Phase VAR Calculation)

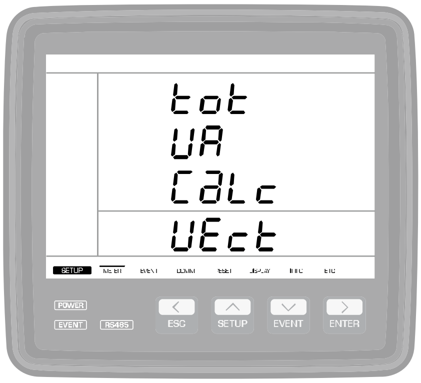
Accura 3300E supports two methods for the calculation of reactive power[Q] and apparent power[S] per phase. One method is to calculate power based on the fundamental frequency components, and the other is to calculate power based on the RMS voltage and RMS current including harmonics. Refer to Chapter 4 for details.



Setup Range	
Fund [Fundamental, default]	Calculation using the fundamental frequency components
Harn [Harmonics]	Calculation using RMS values including harmonics

Calculation of Total Power (tot VA CaLc → Total VA Calculation)

Accura 3300E supports two methods for the calculation of the total power based on the power per phase. One method is to calculate the total power based on the vector sum of the apparent power, and the other is to calculate the total power based on the arithmetic sum of three-phase apparent power vectors. Refer to Chapter 4 for details.



Setup Range	
UEct [Vector, default]	Vector sum method
ScaLa [Scala]	Arithmetic sum method

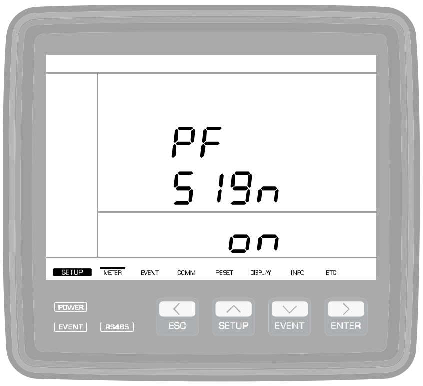
Minimum Measured Current (Min → Minimum, Curr → Current)

A small amount of current can be measured even in no-load condition. It can cause confusion in determining the load state or the ON/OFF state of a breaker. To prevent this, the minimum measured current should be configured, and 0 should be displayed as the current value under the configured value. The current unit is mA, and the minimum measured value is configured based on the value of the secondary side of the external CT. When configuring the minimum measured value as 20 mA(default value) in the Accura 3300E-5A model, any value no more than 1/250 of the rating would be displayed as 0 when it is connected to the CT with a secondary current rating of 5 A.

Setup Range	
5 – 100 mA [default 20]	Accura 3300E-5A Minimum measured current
5 – 20 mA [default 10]	Accura 3300E-1A Minimum measured current

Power Factor Sign (PF → Power Factor, Sign → Sign)

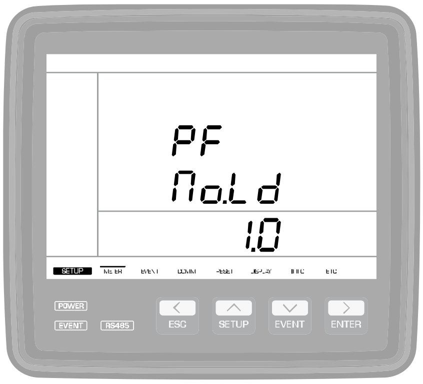
As the power factor is calculated by dividing the active power by the apparent power, the value is positive when energy is received while it is negative when energy is delivered. Refer to Chapter 4 for details. If the power factor sign is set to "Off", the power factor is displayed as an absolute value regardless of whether energy is received or delivered.



Setup Range	
On [ON, default]	The PF sign is displayed.
oFF [OFF]	The PF sign is not displayed.

Power Factor at No Load (PF → Power Factor, No.Ld → No Load)

Power factor cannot be calculated at no load since current is not measured. In this case, you can set up the power factor to be displayed as 1 or 0.



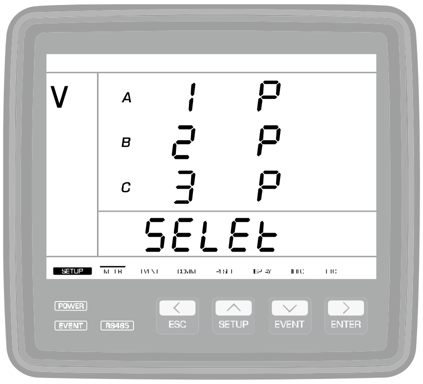
Setup Range	
1.0 [default]	Displays power factor as 1.0 at no load
0.0	Displays power factor as 0.0 at no load

Sign of Reactive Power (Uar → Var, Sign → Sign)

The sign of reactive power is determined according to the calculation method of reactive power or the reactive power is calculated as an absolute value. If you select the calculation method using RMS values including harmonic components, the sign of reactive power is the same as the sign of the reactive power obtained with the calculation method using fundamental frequency components. If "Var Sign" is set to "On", the sign of reactive power is displayed. However, if it is set to "Off", the sign of reactive power is not displayed.

Setup Range	
on[ON, default]	Sign of reactive power is displayed.
oFF[OFF]	Sign of reactive power is not displayed.

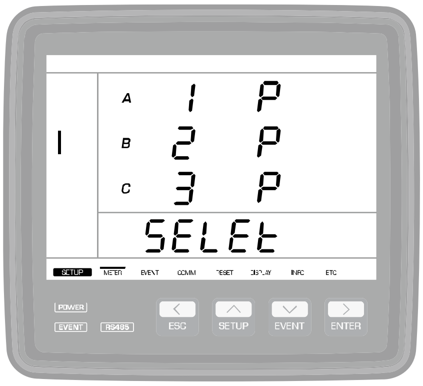
Voltage Phase Selection (SELEt → Select)



Setup Range	
A [d] [1] B [d] C [d] [default 123]	The wiring sequence of voltage phases: 1, 2, and 3 indicates A, B, and C phases are connected to the V1, V2, and V3 voltage input ports, respectively.
P [Positive, default]	Normal voltage wiring
N [Negative]	Reverse voltage wiring

1. d → decimal number

Current Phase Selection (SELEt → Select)



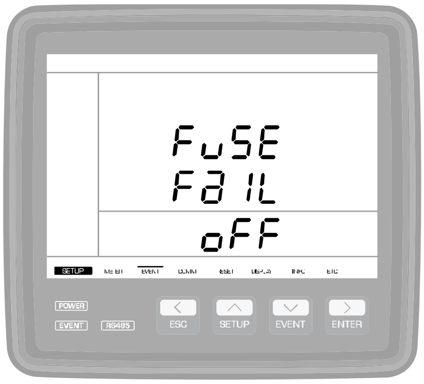
Setup Range	
A [d] [1] B [d] C [d] [default 123]	The wiring sequence of current phases: 1,2, and 3 indicates A, B, and C phases are connected to the current input ports, 1, 2, and 3, respectively.
P [Positive, default]	Normal current flow from the Source(S) to the Load(L)
N [Negative]	Reverse current flow from the Load(L) to the Source(S)

1. d → decimal number

Event Setup

Fuse fail Event (FuSE FaIL → Fuse Fail)

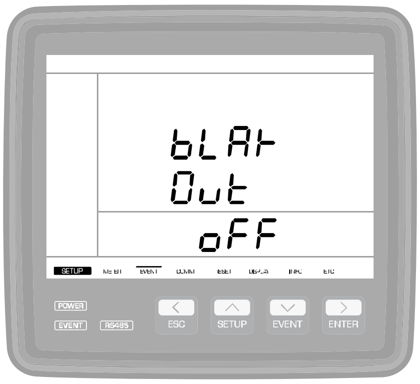
When the voltage value measured on each phase is zero, and the measured current is a non-zero value, it is deemed that a fuse failure occurs on the line where voltage is measured.



Setup Range	
On	Fuse fail event detection enabled
oFF[default]	Fuse fail event detection disabled

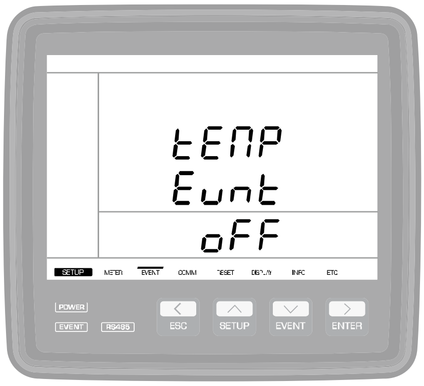
Phase Open & Blackout Event (bLak Out → Blackout)

When the measured voltage and current values for each phase are zero, it is deemed that a phase open event occurs. When the measured voltage and current values for all phases are zero, it is deemed that a blackout event occurs.



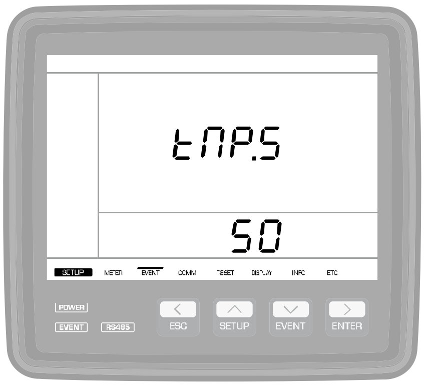
Setup Range	
On	Phase open and blackout event detection enabled
oFF[default]	Phase open and blackout event detection disabled

Over-Temperature Event (tEnP Eunt → Temperature Event)



Setup Range	
On	Over-temperature event detection enabled
oFF [default]	Over-temperature event detection disabled

Start Threshold of Over-Temperature Event (tnP.S → Temperature Start)



Setup Range	
20 – 200 ℃ [default 50]	Start threshold of an over-temperature event

End Threshold of Over-Temperature Event (tnP.E → Temperature End)

Setup Range	
0 – (start threshold -1) ℃ [default 48]	End threshold of an over-temperature event

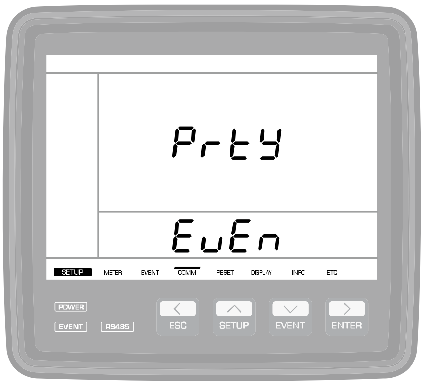
RS-485 Communication Setup

Communication Speed (bAud → Baud rate)

Setup Range	
1200	1,200 bps
2400	2,400 bps
4800	4,800 bps
9600 [default]	9,600 bps
19200	19,200 bps
38400	38,400 bps
57600	57,600 bps
115200 [1]	115,200 bps

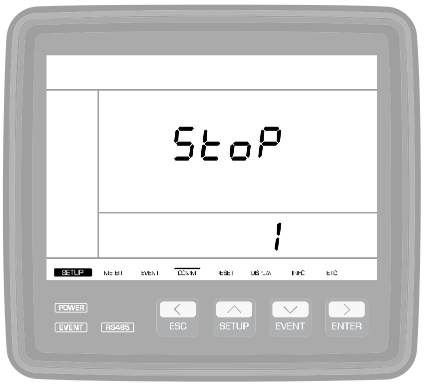
1. At 115,200 bps, a master can communicate with up to 16 slave devices, and the maximum distance between the master and the slave is 600 m.

Parity Bit (Prty → Parity Bit)



Setup Range	
EuEn [default]	Even parity
nonE	No parity
odd	Odd parity

Stop Bit (StoP→ Stop Bit)



Setup Range	
1 [default]	1 stop bit
2	2 stop bits

Address (Addr → Address: RS-485 Communication Address)

In order to read measurement data via communication, the device address should be set to a valid value not zero.

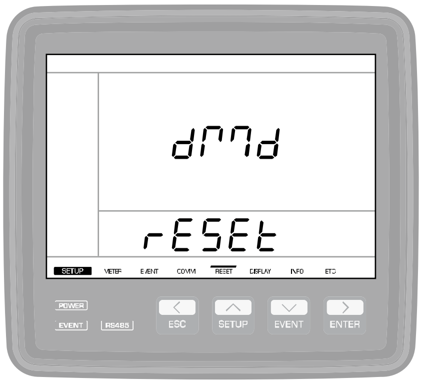
Setup Range	
[d][d][d] [1]	0, 1 – 247
[default 0]	

1. d → decimal number

Reset Setup

On all reset screens, long-pressing the **ENTER** button makes the "rESEt" text appearing in the fourth line blink. If the "rESEt" text stops blinking by long-pressing the **ENTER** button again, resetting the corresponding parameter is completed.

Demand Reset (dMd → *Demand*)



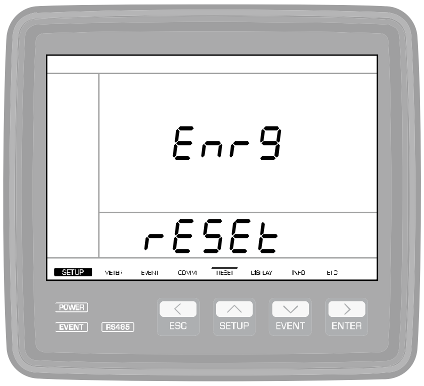
Setup Range	
rESEt [Reset]	Resets demand values

Maximum/Minimum Reset (MaH → Maximum, Min → Minimum)

The maximum and minimum demand values are reset with the reset function for the maximum/minimum values.

Setup Range	
rESEt [Reset]	Resets the maximum/minimum values

Energy Reset (Enrg → Energy)



Setup Range	
rESEt (Reset)	Resets energy[kWh, kVARh, kVAh] values

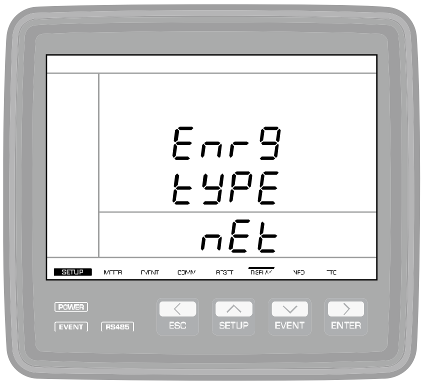
All Reset

Setup Range	
rESEt (Reset)	Resets all parameters (demand, maximum/minimum, energy)

DISPLAY Setup

Energy Type(Enrg → Energy, tyPE → Type)

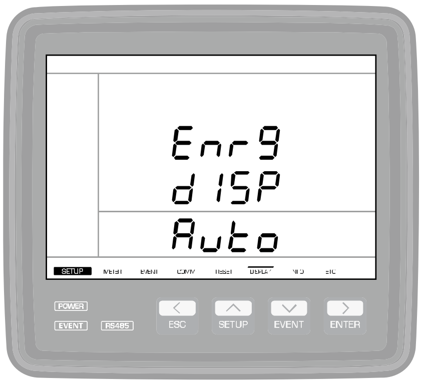
In the display mode, select the energy type to be displayed on the "Voltage/Current/Power/Energy (V·I·P·E)" screen.



Setup Range	
nEt [Net, default]	Net energy (received energy – delivered energy)
rEc [Receive]	Received energy
dEL [Deliver]	Delivered energy
tot [Total]	Total energy (received + delivered energy)

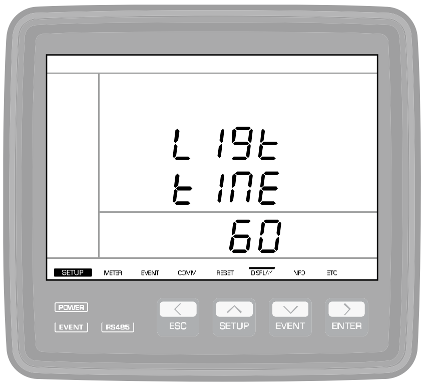
Display of Energy Values (Enrg → Energy, dISP→ Display)

Select the five digits of the energy value to be displayed on the "Voltage/Current/Power/Energy (V·I·P·E)" screen.



Setup Range	
Auto [default]	If the energy value is small, lower digit numbers are displayed. As the energy value gets greater, higher digit numbers are displayed.
Low [Low-Fixed]	Five lower-order digit numbers are displayed.

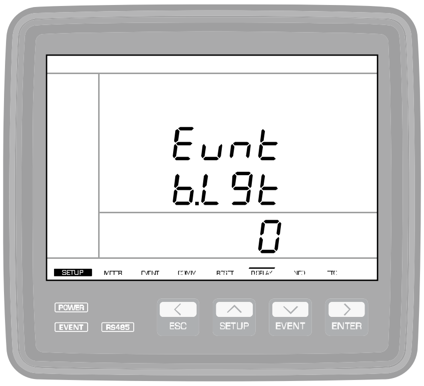
LCD Backlight Timeout (LIgt → Light, tInE → Time)



Setup Range
10 – 300 sec [default 60]

In order to save power consumption, the brightness of the LCD backlight changes to the configured low brightness value after the button is last pressed and the configured time passes.

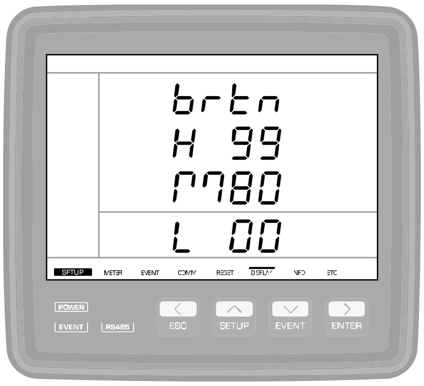
LCD Backlight Timeout When Events Occur (Eunt → Event, b.Lgt → Backlight)



Setup Range	
0 – 9,999 min [default 0]	0: blinks for 5 seconds
Пo.oFF [No.OFF]	

When an event occurs, the LCD backlight blinks during the set timeout period. However, if it is set to the default value, it blinks for 5 seconds. If it is set to "No.OFF", the blinking of the backlight can be reset by checking the event log.

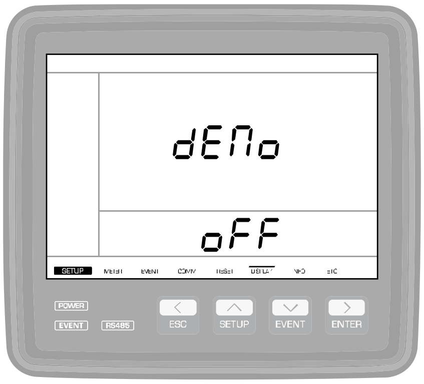
LCD Brightness (brtn → Brightness)



Setup Range	
H.[d][d] [default 99]	High: 80 – 99 %
M.[d][d] [default 80]	Medium: 50 – 80 %
L.[d][d] [default 0]	Low: 0 – 20 %

High	Backlight brightness when users view or configure measured values by pressing the button.
Medium	The backlight brightness changes from the set High value to the set Medium value after the button is last pressed and a certain time passes.
Low	After changing to the set Medium value, brightness is gradually diminished to the set Low value.

Demo (dEno → Demo)



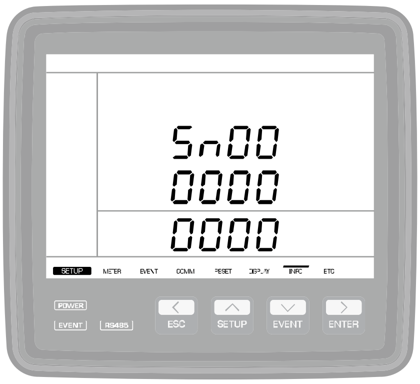
Setup Range	
oFF (OFF, default)	Demo mode disabled
baL (Balance)	Demo mode enabled (3-phase balance)
UnbaL (Unbalance)	Demo mode enabled (3-phase unbalance)

The following table shows the features of each type of the demo modes available according to the demo mode settings. The size of current is calculated by multiplying the CT ratio by 0.2 A (2 A when the default CT ratio is 10), and the current lags the voltage by 60 degrees.

Demo Type	Purpose	Feature	Voltage
Balance	Provides a balanced 3 phase condition	Phase angle of the voltage per phase: 0°, -120°, 120°	When the PT ratio is 1, it provides the line-to-line voltage of 380 V. When the PT ratio changes, it provides the voltage proportional to the PT ratio.
Unbalance	Provides an unbalanced 3 phase condition	Phase angle of the voltage per phase: 0°, -128.4°, 111.6°	

INFO/ETC Setup

Product Serial Number (Sn → Serial Number)



Setup Range	
[d][d][d][d][d][d] [d][d][d][d]	10-digit serial number of the product

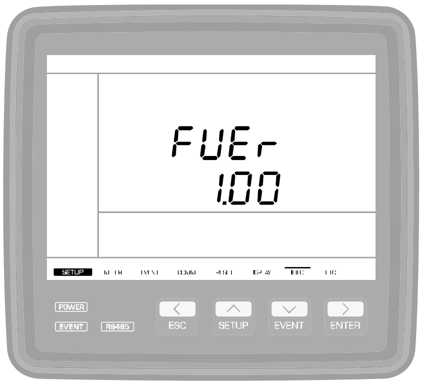
1. d → decimal number

Hardware Version (HUEr → Hardware Version)

Setup Range	
[d].[d][d] [1]	Hardware revision of the product
Ct_[d]A	CT type (Accura 3300E-5A/1A)

1. d → decimal number

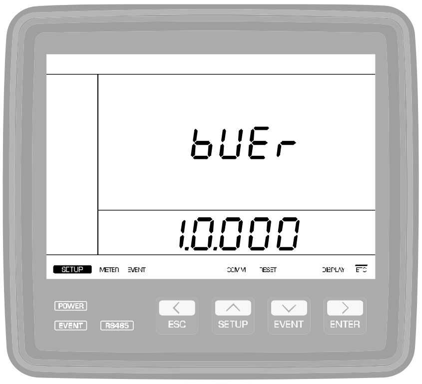
Firmware Version (FUEr → Firmware Version)



Setup Range	
[d].[d][d] [1]	Firmware version of the product

1. d → decimal number

Bootloader Version (bUEr → Bootloader Version)

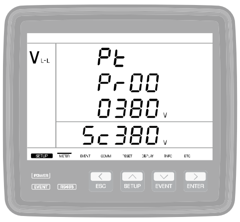


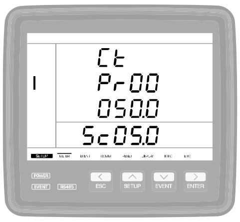
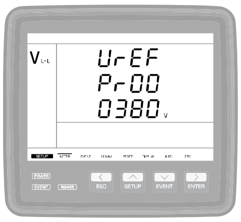
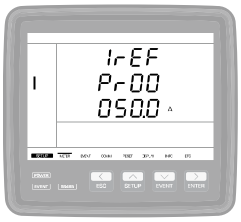
Setup Range	
(d).(d).(d)(d)(d) [1]	Bootloader version of the product

1. d → decimal number

Necessary Steps for Device Setup

For normal operation of Accura 3300E, the following steps for device setup should be taken.

<table>
<tr><th colspan="2">Connection Mode (Conn → Connection]</th></tr>
<tr><td>
</td><td>The connection mode is set to 3-phase 4-wire by default. Thus, in 3P4W mode, skip the following steps(2, 3, 4, 5). In order to change the wiring mode, follow the steps below(an example of 3P3W connection).
① Long press[1] the SETUP button → Enter the setup mode.
② Move to the "METER" menu by pressing the Left[<] or Right[>] arrow button in the setup mode. "METER" is the first menu that appears in the setup mode, and the first screen appearing in the "METER" menu is connection[Conn] mode setup screen.
③ Long press the ENTER button on the [Conn] setup screen→ The text that reads "3P4U" blinks.
④ Change the connection mode from "3P4U(3P4W)" to "3P3U(3P3W)" with the Up(^) or Down(v) arrow button.
⑤ Long press the ENTER button → The setup item stops blinking when it is set to "3P3U(3P3W)".</td></tr>
<tr><th colspan="2">PT Rating (Pt → PT)</th></tr>
<tr><td>
</td><td>① Move to the "PT rating[Pt]" setup screen by pressing the Down[v] arrow button on the "Conn" mode setup screen.
A. Pr[Primary] → PT primary voltage rating [2]
B. Sc[Secondary] → PT secondary voltage rating
The ratio of the primary and secondary voltages is used as the conversion ratio for the primary voltage. Thus, the primary and secondary line-to-line voltage or line-to-neutral voltage can be entered.
② Long press the ENTER button → Only configurable numbers blink.
Select the number to set up by pressing Up[^] or Down[v] button.
③ Press the Left(<) and Right(>) arrow buttons to move to the next digit to select. Keep pressing the Right(>) arrow button to move to the "Sc(Secondary)" setup item.
④ Repeat steps ③ and ④ until the selection of all digits is completed.
⑤ Long press the ENTER button → Setup for all numbers is completed.</td></tr>
</table>

CT Rating (Ct → CT)	
	① Move to the "CT rating(Ct)" setup screen by pressing the **Down[v]** arrow button on the "PT rating(Pt)" setup screen. A. Pr(Primary) → CT primary current rating [3] B. Sc(Secondary) → CT secondary current rating ② Long press the **ENTER** button. → Only configurable numbers blink. ③ Select the number to set up by pressing the **Up[^]** or **Down[v]** arrow button. ④ Press the **Left(<)** and **Right(>)** arrow buttons to move to the next digit to select. ⑤ Keep pressing the **Right(>)** arrow button to move to the "Sc(Secondary)" setup item. ⑥ Repeat steps ③ and ④ until the selection of all digits is completed. ⑦ Long press the **ENTER** button. → Setup for all numbers is completed.
Reference Voltage(UrEF)	
	① Move to the "Reference Voltage(UrEF)" setup screen by pressing the **Down[v]** arrow button on the "CT rating[Ct]" setup screen. Pr(Primary) → Configure the reference voltage. [4] The primary line-to-line voltage value should be entered all the time regardless of the connection mode. ② Long press the **ENTER** button. → Only configurable numbers blink. ③ Select the number to set up by pressing the **Up[^]** or **Down[v]** arrow button. ④ Press the **Left(<)** and **Right(>)** arrow buttons to move to the next digit to select. ⑤ Repeat steps ③ and ④ until the selection of all digits is completed. ⑥ Long press the **ENTER** button. → Setup for all numbers is completed.
Reference Current(IrEF)	
	① Move to the "Reference Current(UrEF)" setup screen by pressing the **Down[v]** arrow button on the "Reference Voltage(UrEF)" setup screen. Pr(Primary) → Configure the reference current. [4] ② Long press the **ENTER** button. → Only configurable numbers blink. ③ Select the number to set up by pressing the **Up[^]** or **Down[v]** arrow button. ④ Press the **Left(<)** and **Right(>)** arrow buttons to move to the next digit to select. ⑤ Repeat steps ③ and ④ until the selection of all digits is completed. ⑥ Long press the **ENTER** button.→ Setup for all numbers is completed. ⑦ Long press the **ESC** button. → Move to the display mode.

1. Press the button for 1 second. The beep sound occurs in 1 second.
2. The primary and secondary voltage ratings of the PT are used to determine the ratio only for measuring the voltage.
3. The primary and secondary current ratings of the CT are used to determine the ratio only for measuring the current.
4. It is used as the reference value for the bar graph displayed on the screen

Chapter 4 Measurements

Accura 3300E detects the frequency of the power source automatically. For frequencies higher than 55 Hz, measurements are performed at 60 Hz, and for those lower than 55 Hz, measurements are performed at 50 Hz. It performs gapless measurements of voltage and current signals and generates measurement data for 0.2 seconds. By averaging five sets of the data measured at 0.2-second intervals, it generates measurement data for 1 second.

Measured Signals	Sampling Rate
Voltage	512 samples/cycle
Current	256 samples/cycle

Frequency	Cycle	0.2 sec
60 Hz	1 cycle (16.7 ms)	12 cycles
50 Hz	1 cycle (20.0 ms)	10 cycles

General Measurement Parameters

General measurement parameters are shown in the table below. Some parameters are displayed on the screen of Accura 3300E and provided via communication. There are also some parameters provided only via communications.

The LCD screen on the front of the device provides measurement values obtained at 1-second aggregation intervals based on the measurement values obtained at 0.2-second intervals.

The maximum and minimum values the device provides after resetting the values are stored in non-volatile memory. Thus, the data are maintained even when Accura 3300E is turned off and then on. Peak demand values are reset to zero by resetting the maximum/minimum values not demand values.

Parameters	Subparameters	Phase	Average/ Total	After Resetting		Unit
				Max.	Min.	
Voltage	Line-to-Neutral	▣	▣	▣	▣	V
	Line-to-Line	▣	▣	▣	▣	V
	Fundamental LN/LL voltage [1]	▲	▲			V
	THD [2]	▣		▲		%
	Phasor	▲				V
	Line-to-Neutral unbalance (NEMA MG1)		▲	▲		%
	Line-to-Line unbalance (NEMA MG1)		▲	▲		%
	Zero-sequence unbalance		▲	▲		%
	Negative-sequence unbalance		▲	▲		%
	Residual voltage [6]		▲	▲	▲	V
	Frequency		▣	▲	▲	Hz

Current	Current	▣	▣	▣	▣	A
	Fundamental	▣	▣			A
	THD [2]	▣		▲		%
	TDD [3]	▣		▲		%
	Phasor	▲				A
	Unbalance (NEMA MG1)		▣	▲		%
	Zero-sequence unbalance		▣	▲		%
	Negative-sequence unbalance		▣	▲		%
	Crest Factor [4]	▲		▲		-
	K-Factor [5]	▲		▲		-
	Residual current [7]		▣	▲	▲	A
Power	Active power	▣	▣	▣	▣	kW
	Reactive power	▣	▣	▣	▲	kVAR
	Apparent power	▣	▣	▣	▲	kVA
Power Factor	Power Factor	▣	▣	▲	▲	-
	Phase Lead/Lag	▣	▣	▲	▲	-
Energy	Received Active Energy	▲	▣			kWh
	Delivered Active Energy	▲	▣			kWh
	Sum Active Energy	▲	▣			kWh
	Net Active Energy	▲	▣			kWh
	Positive Reactive Energy	▲	▲			kVARh
	Negative Reactive Energy	▲	▲			kVARh
	Sum Reactive Energy	▲	▲			kVARh
	Net Reactive Energy	▲	▲			kVARh
	Apparent Energy	▲	▲			kVAh
Demand	Demand Current/Peak Demand Current	▣	▣	▣		A
	Predicted Demand Current		▲			A
	Demand Active Power/Peak Demand Active Power	▣	▣	▣		kW
	Predicted Demand Active Power		▲			kW
	Demand Reactive Power/Peak Demand Reactive Power	▲	▲	▲		KVAR
	Predicted Demand Reactive Power		▲			KVAR
	Demand Apparent Power/Peak Demand Apparent Power	▲	▲	▲		KVA
	Predicted Demand Apparent Power		▲			KVA
Harmonics	Magnitude of Voltage Harmonics (DC, 1st, 2nd – 50th)	▲				V
	Magnitude of Current Harmonics (DC, 1st, 2nd - 50th)	▲				A

Instantaneous waveform	2 cycles of voltage waveforms (64 samples/cycle)	▲				V
	2 cycles of current waveforms (64 samples/cycle)	▲				A

1. The fundamental frequency components of line-to-neutral voltage and line-to-line voltage are provided in 3P4W mode and in 3P3W mode, respectively.
2. THD(Total Harmonic Distortion), Voltage THD : $\frac{\sqrt{\sum_{k=2}^{50} V_k^2}}{V_1}$, Current THD : $\frac{\sqrt{\sum_{k=2}^{50} I_k^2}}{I_1}$
3. TDD(Total Demand Distortion) : $\frac{\sqrt{\sum_{k=2}^{50} I_k^2}}{I_L}$

where I_L can be set to the peak demand current(default) or the rated current(only by communication).

4. Crest factor : $\frac{V_{peak}}{V_{rms}}$
5. K-factor : $\frac{\sum_{k=1}^{50} (k\ I_k)^2}{\sum_{k=1}^{50} I_k^2}$
6. The residual voltage is the RMS voltage value obtained by summing the three-phase voltages in the time domain.
7. The residual current is three times the zero-sequence current value of fundamental components. When harmonics are absent, the residual current equals the RMS current obtained by summing the three-phase currents in the time domain.

Measurement Event Parameters

The following table shows the parameters of the measurement events provided via Accura 3300E and communication. You can check detailed information on the last 50 events through event logs. In the event mode, the "Event" segment in the top menu bar lights up to indicate the event mode.

▣ Displayed on Accura 3300E and Provided Via Communication **▲ Available Only Via Communication**

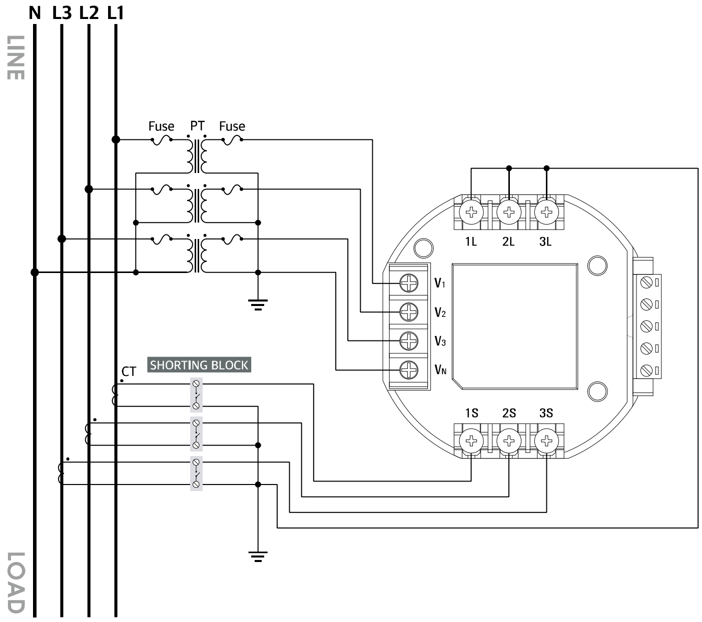
Event Parameters	Per Phase	Total	Start Event	End Event
			Start Value	End Value
Phase Open	▣		-	-
Fuse Fail	▣		-	-
Blackout		▣		-
Over-temperature		▣	▣	▣

Measurements according to Voltage Wiring Modes

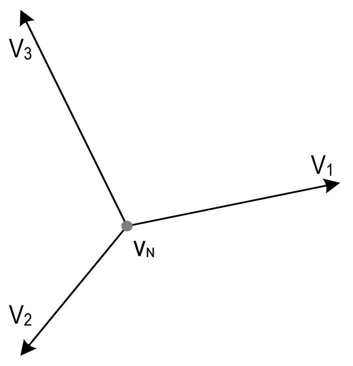
The following wiring diagrams show how to measure voltage, current and power according to the voltage wiring mode.

3-Phase 4-Wire Connection

Fig 4.1 Connection with 3 PTs and 3 CTs, Wiring Mode = 3P4U (3P4W Connection)



Voltage Measurement

Vector Diagram	Item	Description
 V_N : Voltage of connected N terminal	Line-to-neutral voltage sensing	▶ V_1, V_2, and V_3 sensing based on the voltage at the wired N terminal (actual line-to- neutral voltage) V_A = V1 V_B = V2 V_C = V3
	Calculation of line-to line voltage	$V_{AB} = V_A - V_B$ = V1 – V2 $V_{BC} = V_B - V_C$ = V2 – V3 $V_{CA} = V_C - V_A$ = V3 – V1

Available Measurement Values

Item	L-N Voltage [rms]	L-L Voltage [rms]	Current [rms]	Active Power
A Phase	V_A	V_{AB}	I_A (I1)	$P_A = V_A * I_A * PF_A$
B Phase	V_B	V_{BC}	I_B (I2)	$P_B = V_B * I_B * PF_B$
C Phase	V_C	V_{CA}	I_C (I3)	$P_C = V_C * I_C * PF_C$
Average/Total	$(V_A+V_B+V_C)/3$	$(V_{AB}+V_{BC}+V_{CA})/3$	$(I_A+I_B+I_C)/3$	$P_A+P_B+P_C$

3-Phase 3-Wire Connection

Fig 4.2 Connection with 2 PTs and 3 CTs, Wiring Mode = 3P3U (3P3W Connection)

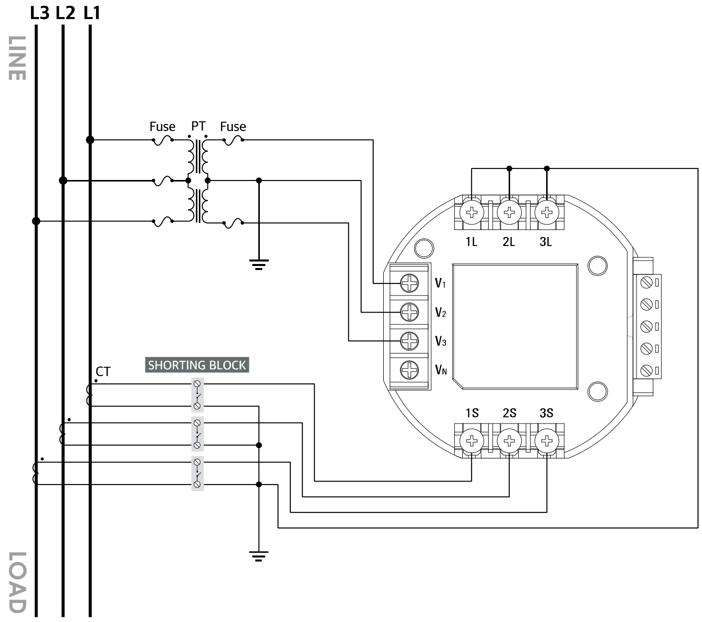
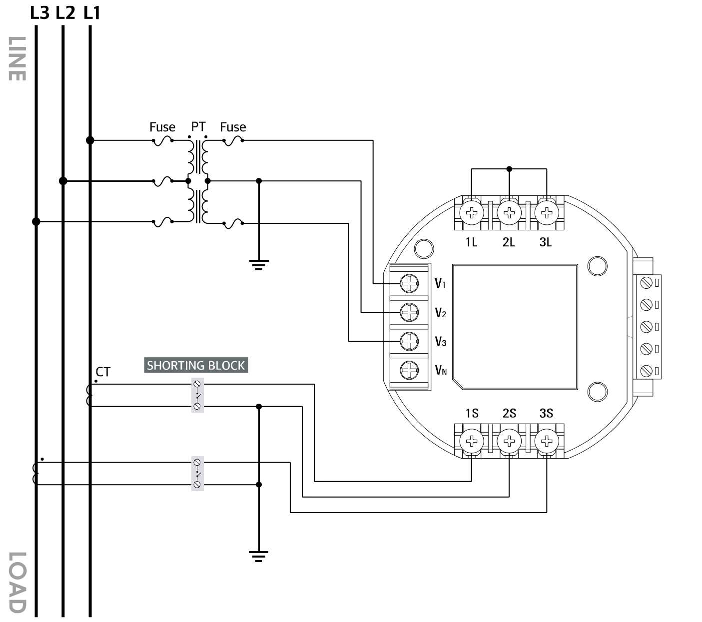
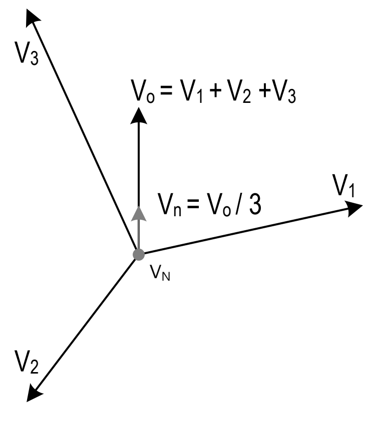
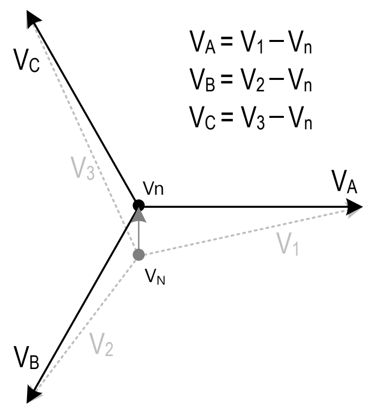


Fig 4.3 Connection with 2 PTs and 2 CTs, Wiring Mode = 3P3U (3P3W Connection)



You can use 2 or 3 PTs(Potential Transformers) in 3-phase 3-wire mode. Its installation cost differs according to the number of PTs. However, regardless of how many PTs are used, Accura 3300E measures voltage signals in the same way. By using 3 CTs, you can measure the residual current by summing three-phase current values.

<table>
<tr><th>Vector Diagram</th><th>Item</th><th>Description</th></tr>
<tr><td rowspan="3">

V_N : Voltage at the unwired N terminal</td><td>Line-to-neutral voltage sensing</td><td>▶ Arbitrary line-to-neutral voltage(V1, V2, V3) sensing based on the voltage at the unwired N terminal</td></tr>
<tr><td>Calculation of residual voltage</td><td>▶ Calculates residual voltage based on the arbitrary line-to-neutral voltages sensed
Vo = V1 + V2 + V3</td></tr>
<tr><td>Calculation of virtual neutral voltage</td><td>▶ Calculates virtual neutral voltage by dividing residual voltage by 3
Vn = Vo/3
▶ A newly calculated virtual center point is the center of mass of the triangle consisting of V1, V2 and V3.</td></tr>
<tr><td rowspan="2">

V_n : Virtual neutral voltage</td><td>Calculation of virtual line-to-neutral voltage</td><td>▶ Calculates virtual line-to-neutral voltage based on the virtual neutral voltage
V_A = V1 – Vn
V_B = V2 – Vn
V_C = V3 – Vn

The residual voltage value of virtual line-to-neutral voltage is 0.</td></tr>
<tr><td>Calculation of line-to-line voltage</td><td>$V_{AB} = V_A - V_B$ = V1 – V2
$V_{BC} = V_B - V_C$ = V2 – V3
$V_{CA} = V_C - V_A$ = V3 – V1</td></tr>
</table>

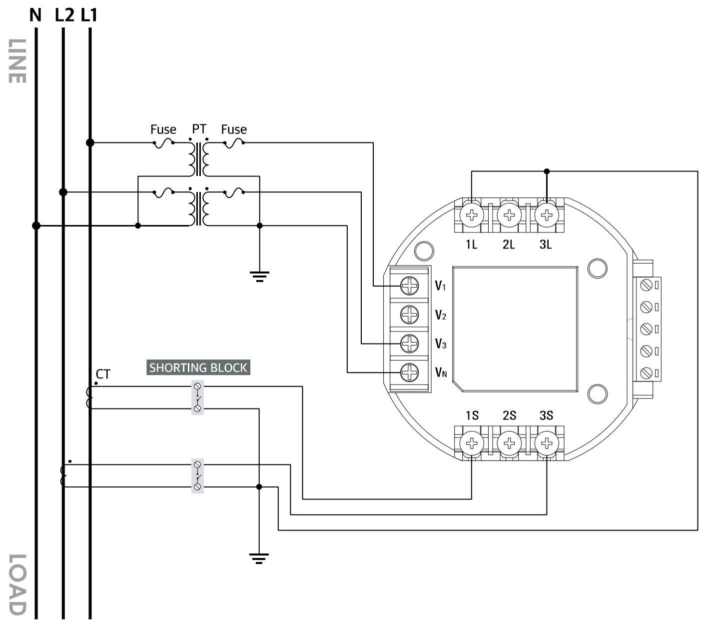
Available Measurement Values

Item	L-N Voltage [rms]	L-L Voltage [rms]	Current [rms]	Active Power
A Phase	V_A	V_{AB}	I_A (I1)	$P_A = V_A * I_A * PF_A$
B Phase	V_B	V_{BC}	I_B (I2)	$P_B = V_B * I_B * PF_B$
C Phase	V_C	V_{CA}	I_C (I3)	$P_C = V_C * I_C * PF_C$
Average/Total	$(V_A+V_B+V_C)/3$	$(V_{AB}+V_{BC}+V_{CA})/3$	$(I_A+I_B+I_C)/3$	$P_A+P_B+P_C$

There is no actual neutral point in the 3-phase 3-wire voltage connection mode. Thus, you can find out only the line-to-line voltage and the total power, and the data on the actual voltage and the power for each phase cannot be obtained. However, by using a virtual neutral point where the residual voltage value of the three phase voltages becomes zero, voltage and power values for each phase can be calculated.

1-Phase 3-Wire Connection

Fig 4.4 Connection with 2 PTs and 2 CTs, Wiring Mode = 1P3U (1P3W Connection)



Voltage Measurement

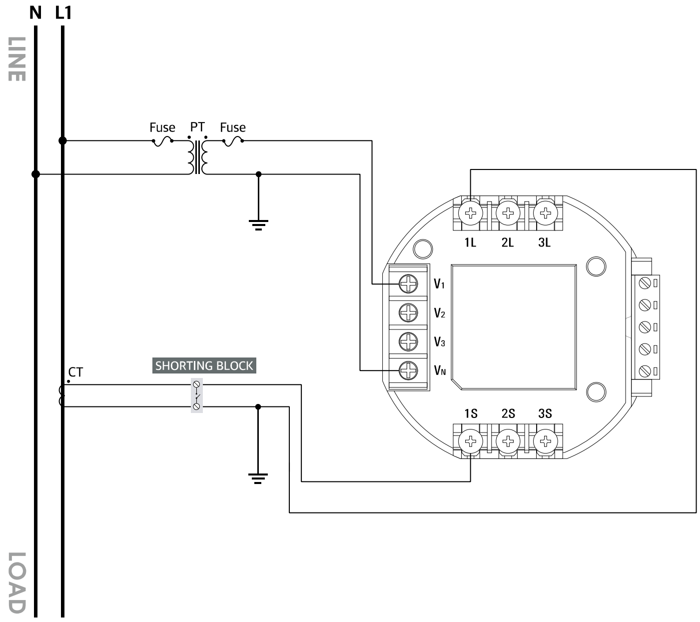
Vector Diagram	Item	Description
$V_3 \leftarrow V_N \rightarrow V_1$ V_N: Voltage at the unwired N terminal	Line-to-neutral voltage sensing	▶ V1 and V3(line-to-neutral voltage) sensing based on the voltage at the wired N terminal (actual line-to-neutral voltage) V_A = V1 V_B = 0 (The value becomes 0 in 1P3W mode.) V_C = V3
	Calculation of line-to-line voltage	V_{AB} = 0 (The value becomes 0 in 1P3W mode.) V_{BC} = 0 (The value becomes 0 in 1P3W mode.) $V_{CA} = V_C - V_A$ = V3 – V1

Available Measurement Values

Item	L-N Voltage [rms]	L-L Voltage [rms]	Current [rms]	Active Power
Phase A	V_A	0	I_A (I1)	$P_A = V_A * I_A * PF_A$
Phase B	0	0	0	0
Phase C	V_C	V_{CA}	I_C (I3)	$P_C = V_C * I_C * PF_C$
Average/Total	$(V_A+V_C)/2$	V_{CA}	$(I_A+I_C)/2$	P_A+P_C

1-Phase 2-Wire Connection

Fig 4.5 Connection with 1 PT and 1 CT, Wiring Mode = 1P2U (1P2W Connection)



Voltage Measurement

Vector Diagram	Item	Description
V_N → V_1 V_N: Voltage at the unwired N terminal	Line-to-neutral voltage sensing	▶ V1 sensing based on the voltage at the wired N terminal(actual line-to-neutral voltage) V_A = V1 V_B = 0 (The value becomes 0 in 1P2W mode.) V_C = 0 (The value becomes 0 in 1P2W mode.)
	Calculation of line-to-line voltage	V_{AB} = V_A V_{BC} = 0 (The value becomes 0 in 1P2W mode.) V_{CA} = 0 (The value becomes 0 in 1P2W mode.)

Available Measurement Values

Item	L-N Voltage [rms]	L-L Voltage [rms]	Current [rms]	Active Power
A Phase	V_A	V_A	I_A (I1)	$P_A = V_A * I_A * PF_A$
B Phase	0	0	0	0
C Phase	0	0	0	0
Average/Total	V_A	V_A	I_A	P_A

Measurement of Power Per Phase

There are two methods for calculation of reactive[Q] and apparent power[S] per phase.

(1) Calculation with fundamental frequency components: Reactive energy is calculated using the fundamental voltage and current values.

(2) Calculation with RMS values, including harmonics: Reactive energy is calculated using the RMS voltage and current values.

Reactive power and apparent power values can vary according to the method of calculating reactive power. As the apparent power value changes, the power factor also changes.

Calculation with Fundamental Frequency Components

Active power for each phase is calculated by multiplying voltage by current.

$$P_a = \frac{1}{T}\int_0^T V_a(t) I_a(t)dt\,,\quad P_b = \frac{1}{T}\int_0^T V_b(t) I_b(t)dt\,,\quad P_c = \frac{1}{T}\int_0^T V_c(t) I_c(t)dt$$

The reactive power for each phase is calculated by shifting the phase of the voltage by 90 degrees with respect to the fundamental frequency.

$$Q_a = \frac{1}{T}\int_0^T V_a(t-\tfrac{T}{4}) I_a(t)dt\,,\quad Q_b = \frac{1}{T}\int_0^T V_b(t-\tfrac{T}{4}) I_b(t)dt\,,\quad Q_c = \frac{1}{T}\int_0^T V_c(t-\tfrac{T}{4}) I_c(t)dt$$

The apparent power for each phase is calculated based on the magnitude of the vector sum of the active power and the reactive power.

$$S_a = \sqrt{P_a^2 + Q_a^2}\,,\quad S_b = \sqrt{P_b^2 + Q_b^2}\,,\quad S_c = \sqrt{P_c^2 + Q_c^2}$$

Calculation with RMS Values, including Harmonics

The active power for each phase is calculated by multiplying voltage by current.

$$P_a = \frac{1}{T}\int_0^T V_a(t) I_a(t)dt\,,\quad P_b = \frac{1}{T}\int_0^T V_b(t) I_b(t)dt\,,\quad P_c = \frac{1}{T}\int_0^T V_c(t) I_c(t)dt$$

The apparent power for each phase is calculated by multiplying the RMS voltage by the RMS current. The RMS values of voltage and current contain harmonic components.

$$V_{k,rms} = \sqrt{\frac{1}{T}\int_0^T v_k^2(t)\,dt}\,,\qquad I_{k,rms} = \sqrt{\frac{1}{T}\int_0^T i_k^2(t)\,dt}$$

$$S_a = V_{a,rms}\,I_{a,rms}\,,\qquad S_b = V_{b,rms}\,I_{b,rms}\,,\quad S_c = V_{c,rms}\,I_{c,rms}$$

The magnitude of the reactive power for each phase is calculated based on the magnitude of the vector difference between the apparent power and the active power. The sign of the reactive power is the same as the sign of the reactive power calculated with fundamental frequency components.

$$Q_a = \pm\sqrt{S_a^2 - P_a^2}\qquad Q_b = \pm\sqrt{S_b^2 - P_b^2}\qquad Q_c = \pm\sqrt{S_c^2 - P_c^2}$$

Three-Phase Power Measurement

There are two methods for calculation of the total reactive[Qt] and apparent power[St] for three phases.

(1) Vector sum: The total apparent power is calculated based on the vector sum of three-phase apparent power.

(2) Arithmetic sum: The total apparent power is calculated based on the arithmetic sum of three-phase apparent power.

The total apparent and reactive power values can vary according to the method of calculating the total apparent power. As the total apparent power value changes, the total power factor(PF) also changes.

Vector Sum Calculation

Users can calculate the total active power, reactive power and apparent power based on the vector sum of three-phase power vectors

The active power of each phase is summed.

$$P_t = P_a + P_b + P_c$$

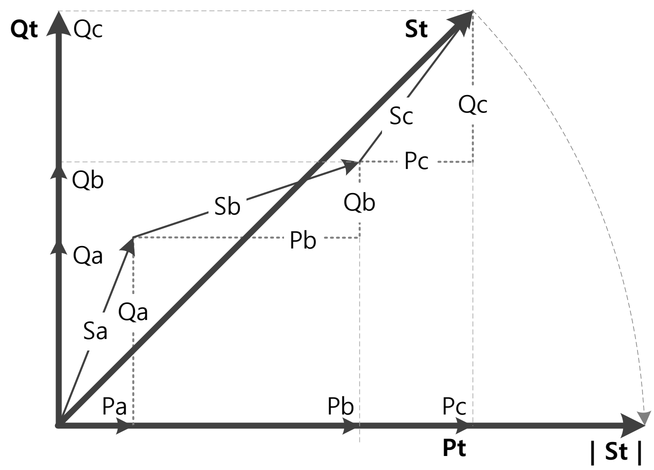
The reactive power of each phase is summed.

$$Q_t = Q_a + Q_b + Q_c$$

The total apparent power is calculated based on the vector sum of the total active power and total reactive power.

$$S_t = \sqrt{P_t^2 + Q_t^2} \quad \textbf{(Vector sum)}$$

Fig 4.6 Vector Sum for the Total Apparent Power



Arithmetic Sum Calculation

The active power of each phase is summed.

$$P_t = P_a + P_b + P_c$$

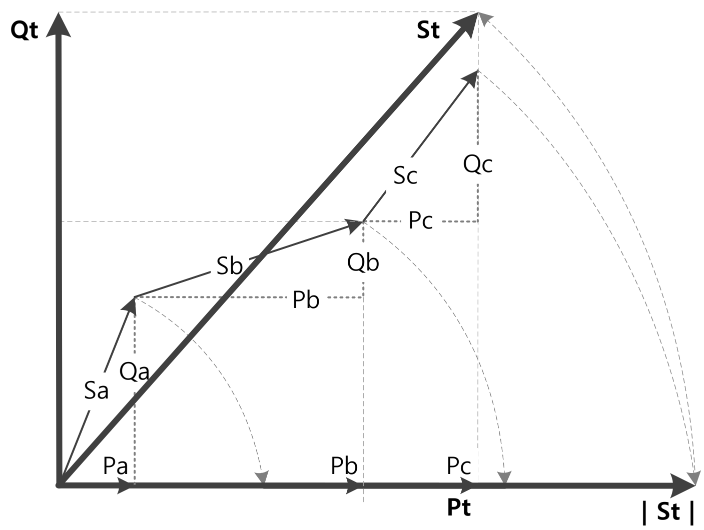
The total apparent power is calculated based on the arithmetic sum of three-phase apparent power vectors.

$$S_t = |S_a| + |S_b| + |S_c| \quad \text{(Arithmetic sum)}$$

The magnitude of the total reactive power is determined by calculating the magnitude of vector difference between the total apparent power and the total active power. The sign of the total reactive power is consistent with the sign of the total reactive power determined by the vector sum.

$$Q_t = \pm\sqrt{S_t^2 - P_t^2}$$

Fig 4.7 Arithmetic Sum for the Total Apparent Power



Power Factor(PF) Measurement

Power factor is calculated as the ratio of the active power to the apparent power. The sign of the power factor can be positive or negative depending on the sign of the active power. If it is positive, it indicates the power is in received status. If it is negative, it indicates the power is in delivered status. If the apparent power is zero, the power factor cannot be calculated normally. In this case, it can be set to 1.0 or 0.0, and the default value is 1.0.

$$PF_a = \frac{P_a}{S_a}, \qquad PF_b = \frac{P_b}{S_b}, \qquad PF_c = \frac{P_c}{S_c}$$

$$PF_t = \frac{P_t}{S_t}$$

The sign of the power factor can be removed through configuration. In this case, the received/delivered status cannot be identified only through the power factor. How to calculate the PF when its sign is removed is as follows.

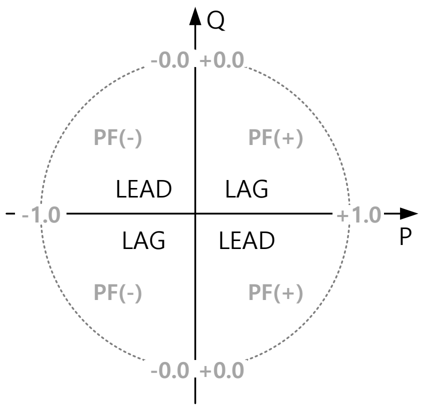
$$PF_a = \left|\frac{P_a}{S_a}\right|, \qquad PF_b = \left|\frac{P_b}{S_b}\right|, \qquad PF_c = \left|\frac{P_c}{S_c}\right|$$

$$PF_t = \left|\frac{P_t}{S_t}\right|$$

Phase Angle Measurement(LEAD/LAG)

The following figure shows the sign of the power factor and the Lead/Lag indicators for the phase angles of voltage and current in 4 quadrants. When the reactive power(Q) is zero, it is in neither Leading nor Lagging status.

Fig 4.8 PF Sign and Phase Angle(Lead/Lag)



Note

If the current leads or lags the voltage by a phase angle, the power factor is displayed as LEAD or LAG on the screen on the front of Accura 3300E. If the reactive power is zero, the information is not displayed.

Demand Measurement

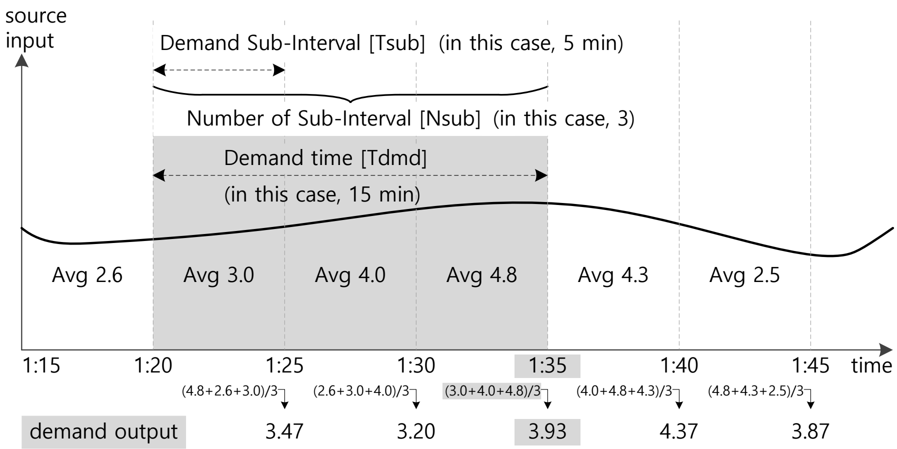
Demand

The demand time(Tdmd) is calculated by multiplying the demand subinterval(Tsub) by the number of demand subintervals(Nsub). The average value of the demand time(Tdmd) is calculated over a sliding demand subinterval. The peak demand is the maximum value obtained among the demand values after resetting the maximum and minimum values, and it can be reset using a command to clear the maximum and minimum values.

Demand time[min] = Demand subinterval x Number of demand subintervals

E.g.) In Korea, when the demand time is 15 minutes, the demand subinterval is 15 minutes, and the number of demand subintervals is 1.

Fig 4.9 Demand Calculation



Thermal Demand

Thermal demand(P_{th}) is calculated every second using the following equation involving the demand value(P) and t value. The thermal demand(P_{th}) value obtained through a first-order LPF(low-pass-filter) is used to calculate predicted demand.

$$P_{th}[n] = \left(1 - \frac{1}{\tau}\right) \cdot P_{th}[n-1] \;+\; \frac{1}{\tau} \cdot P[n]$$

τ refers to the thermal time constant, and it is calculated through the following equation involving the thermal response index.

$$\tau = \frac{100 - (Thermal\ Response\ Index)}{100} \cdot T_{sub} \cdot 60\ [sec]$$

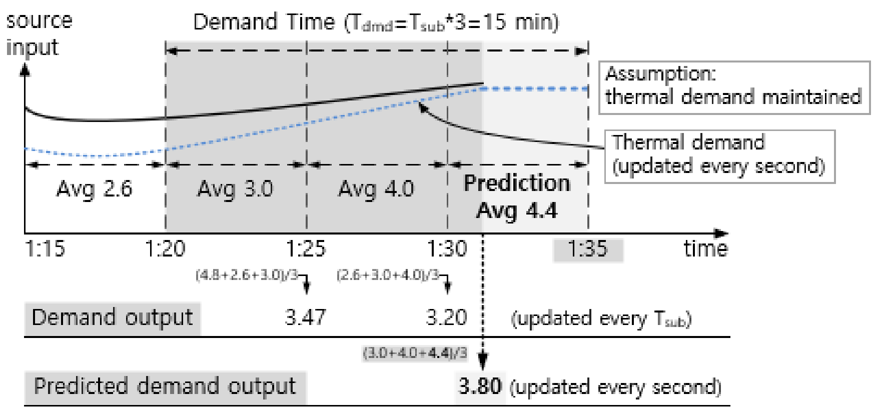
If the thermal time constant τ is less than 1 second, it is calculated as 1 second.

Setup Item	Description	Setup Range	Default
Thermal response index	- Determines the response speed of thermal demand - As a set value is greater, thermal demand features a faster response time. - When the thermal response index is set to 100, the P_{th} value refers to the latest P value.	0 – 100	90

Predicted Demand

Predicted demand refers to predicting the demand at the end of the demand subinterval(Tsub) every second while the subinterval is ongoing. The average value of the current demand subinterval is calculated every second under the assumption that the thermal demand value will remain constant for the duration of the remaining subinterval while the current demand subinterval continues. The predicted demand value at the end of the demand subinterval is calculated using the average value obtained from the current demand subinterval.

Fig 4.10 Predicted Demand Calculation



Demand Sync

Accura 3300E supports two methods for synchronizing the starting time of a demand subinterval.

1) Hourly Auto Sync: refers to automatically synchronizing the starting time of a demand subinterval every hour.
2) Manual Sync: refers to synchronizing the starting time of a demand subinterval at the time users issue a sync command.

For hourly auto sync, synchronization is performed every 60 minutes. Thus, demand calculation is performed only when the demand subinterval is an integer divided by 60(demand subinterval(Tsub)=1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 10 minutes, 12 minutes, 15 minutes, 20 minutes, 30 minutes, and 60 minutes). A sync command for Manual Sync can be issued via communication.

Fig 4.11 Hourly Auto Sync

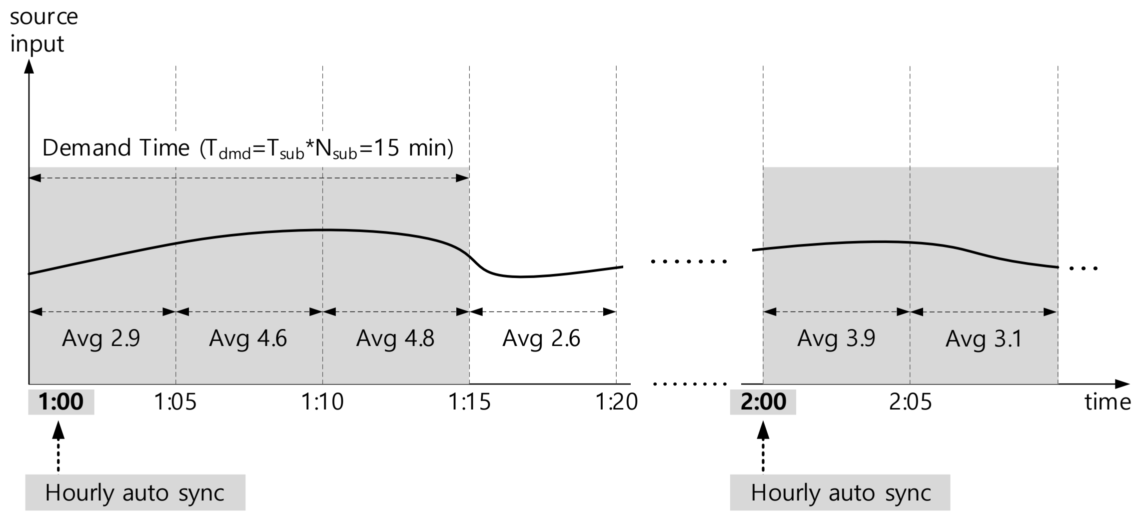
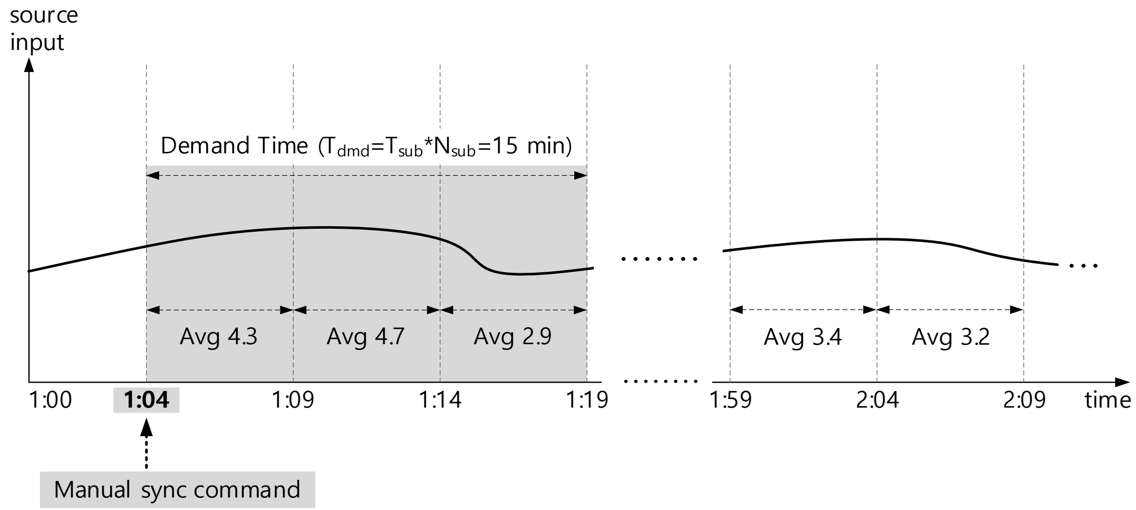


Fig 4.12 Manual Sync



Unbalance(NEMA MG1)

Line-to-Neutral Voltage Unbalance

It indicates the maximum voltage deviation from the average value of three-phase line-to-neutral voltages. Even when the magnitudes of line-to-neutral voltages are the same and voltage unbalance occurs between phases, the unbalance value of the line-to-neutral voltage is zero.

$$Unbalance = \frac{Max(\Delta Vph)}{Vph, avg} \times 100\ [\%]$$

Vph,avg : Average value of the line-to-neutral voltage

$Max(\Delta Vph)$: Maximum voltage deviation of the most deviated phase from the average line-to-neutral voltage.

Line-to-Line Voltage Unbalance

It indicates the maximum deviation degree of line-line voltage from the average of line-line voltages.

$$Unbalance = \frac{Max(\Delta VLL)}{VLL, avg} \times 100\ [\%]$$

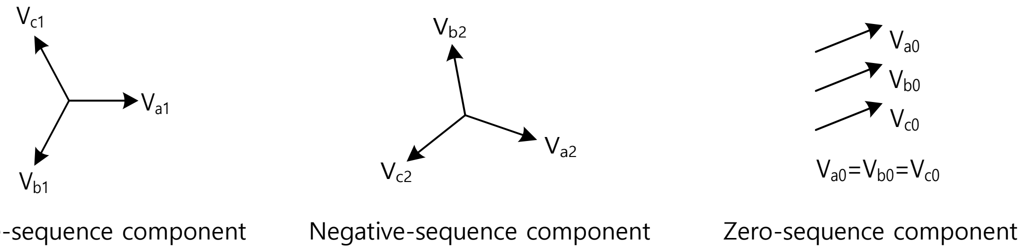
VLL,avg :Average value of the line-to-line voltage

$Max(\Delta VLL)$: Maximum voltage deviation of the most deviated phase from the average line-to-line voltage.

Negative-Sequence/Zero-Sequence Unbalance

The fundamental components of three-phase voltages can be divided into zero-sequence, positive-sequence and negative-sequence components.

Fig 4.13 Symmetrical Components



Zero-sequence component(U0): the component with the same magnitude and phase angle

Positive-sequence component(U1): symmetrical component with the phase sequence of A-B-C

Negative-sequence component(U2): symmetrical component with the phase sequence of A-C-B

Zero-sequence, positive-sequence, and negative-sequence components are calculated with the following equation.

$$\begin{bmatrix} U_0 \\ U_1 \\ U_2 \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \cdot \begin{bmatrix} U_a \\ U_b \\ U_c \end{bmatrix}, \quad where\ a = e^{j \cdot \frac{2\pi}{3}}$$

Negative-Sequence Unbalance

It is the ratio between the magnitudes of negative sequence component and positive sequence component.

$$Negative\ sequence\ unbalance\ ratio = \frac{U_2}{U_1} \times 100\ [\%]$$

Zero-Sequence Unbalance

It is the ratio between the magnitudes of the zero-sequence component and the positive-sequence component.

$$Zero\ sequence\ unbalance = \frac{U_0}{U_1} \times 100\ [\%]$$

Appendix A Specifications

Accura 3300E Digital Power Meter

Voltage Input		
Measuring range (accuracy guaranteed)	AC 35 – 277 V L-N(Line-to-neutral)	
	AC 60 – 480 V L-L (for unearthed Delta & Y systems)	
Minimum measured voltage	AC 5 V L-N(Line-to-neutral), AC 9 V L-L(Line-to-line)	
Frequency range	42 – 65 Hz	
Impedance	10 MΩ/phase	
Burden	0.01 VA/phase @ 220 V	
Withstand voltage	AC 3,000V RMS, 60 Hz for 1minute	
Wiring method	3P4W, 3P3W, 1P3W, 1P2W	
Current Input		
Rating	5 A nominal/10 A full scale 3~	Accura 3300E-5A
	1 A nominal/10 A full scale 3~	Accura 3300E-1A
Measuring range (accuracy guaranteed)	1 %In – 200 %In [1]	Accura 3300E-5A
	5 %In – 200 %In [1]	Accura 3300E-1A
Minimum measured current	5 mA	
Burden	Max. 0.01 VA/phase @ 10 A	
Power Supply		
Power supply voltage [2]	AC 100 – 240 V 50/60 Hz, DC 100 – 300 V, CAT II	
Operating voltage range	0.9 x Us – 1.1 x Us	
Power consumption	3 W	
General		
Weight	510 g	
Operating temperature	-20 – 70 ℃ (-4 – 158 ℉)	
Operating environment	Pollution degree 2	
Safe operating temperature [3]	-20 – 60 ℃ (-4 – 140 ℉)	
Storage temperature	-40 – 85 ℃ (-40 – 185 ℉)	
Operating humidity	5 – 95% (non-condensing)	
Operating altitude	Up to 2,000 m	
Ingress protection against dust and water	IEC 60529 IP54 (Accura 3300E)	

1. The rated current In is 5 A/1 A.
2. The device conforms to UL standards for the tested AC power supply voltage.
3. Compliant with UL 61010-1(3rd edition)

Appendix B Standards

Accuracy	
IEC 62053-22	Static meters for AC active energy(classes 0.1S, 0.2S and 0.5S)
IEC 62053-24	Static meters for fundamental component reactive energy(classes 0.5S, 1S, 1, 2 and 3)
IEC 61557-12	Performance measuring and monitoring devices(PMD)
Power Quality	
IEC 61000-4-30	Power quality measurement methods
Safety	
UL 61010-1 3rd edition	Safety requirements for electrical equipment for measurement, control and laboratory use
EMC	
EN 55011	Conducted emissions
EN 55011	Conducted emissions (asymmetric mode)
EN 55011	Radiated emissions
IEC 61000-3-2	Limits for harmonic current emissions
IEC 61000-3-3	Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems
IEC 61000-4-2	Electrostatic discharge immunity test
IEC 61000-4-3	Radiated, radio-frequency, electromagnetic field immunity test
IEC 61000-4-4	Electrical fast transient/burst immunity test
IEC 61000-4-5	Surge immunity test
IEC 61000-4-6	Immunity to conducted disturbances, induced by radio-frequency fields
IEC 61000-4-8	Power frequency magnetic field immunity test
IEC 61000-4-11	Voltage dips, short interruptions and voltage variations immunity test
Certifications	
UL	UL 61010-1 3rd edition, UL 61010-2-030 2nd edition
CE	EN 55011:2016/A11:2020, EN IEC 61326-1:2021, EN IEC 61326-2-1:2021, EN IEC 61000-3-2:2019/A1:2021, EN 61000-3-3:2013/A2:2021
KC	EN 55011:2016/A11:2020, EN IEC 61326-1:2021, EN IEC 61326-2-1:2021
General	
Warranty period	2 years

IEC 62053-22:2020 Static Meters for AC Active Energy (Class 0.5S)

Voltage Range	Current Range for only transformer operated meters	Power Factor (cosφ)	Class 0.5S
90 %Un – 110 %Un	1 %In – 5 %In	1.0	±1.0 %
	5 %In – Imax	1.0	±0.5 %
	2 %In – 10 %In	0.5 inductive 0.8 capacitive	±1.0 % ±1.0 %
	10 %In – Imax	0.5 inductive 0.8 capacitive	±0.6 % ±0.6 %

1. Un indicates the rated voltage, and In indicates the rated current.

IEC 62053-24:2020 Static Meters for Fundamental Reactive Energy (Class 0.5S)

Voltage Range	Current Range for transformer operated meters	sinφ	Class 0.5S
90 %Un – 110 %Un	1 %In – 5 %In	1.0	±1.0 %
	5 %In – Imax	1.0	±0.5 %
	5 %In – 10 %In	0.5	±1.0 %
	10 %In – Imax	0.5	±0.5 %
		0.25	±1.0 %

1. Un indicates the rated voltage, and In indicates the rated current.

IEC 61557-12:2018 Performance Measuring and Monitoring Devices(PMD)

Active Power and Active Energy

Voltage Range	Current Range		Power Factor (cosφ)	Class 0.5
	PMD Dx	PMD Sx		
80 %Un – 120 %Un	2 %Ib – 10 %Ib	1 %In – 5 %In	1.0	±1.0 %
	10 %Ib – Imax	5 %In – Imax	1.0	±0.5 %
	5 %Ib – 20 %Ib	2 %In – 10 %In	0.5 inductive 0.8 capacitive	±1.0 % ±1.0 %
	20 %Ib – Imax	10 %In – Imax	0.5 inductive 0.8 capacitive	±0.6 % ±0.6 %

Reactive Power and Reactive Energy

Voltage Range	Current Range		sinφ	Class 1
	PMD Dx	PMD Sx		
80 %Un – 120 %Un	5 %Ib – 10 %Ib	2 %In – 5 %In	1.0	±1.25 %
	10 %Ib – Imax	5 %In – Imax	1.0	±1.0 %
	10 %Ib – 20 %Ib	5 %In – 10 %In	0.5	±1.25 %
	20 %Ib – Imax	10 %In – Imax	0.5	±1.0 %
			0.25	±1.25 %

Apparent Power and Apparent Energy

Voltage Range	Current Range		Class 0.5
	PMD Dx	PMD Sx	
80 %Un – 120 %Un	5 %Ib – 10 %Ib	2 %In – 5 %In	±1.0 %
	10 %Ib – Imax	5 %In – Imax	±0.5 %

Frequency

Voltage Range	Current Range		Frequency	Class 0.02
	PMD Dx	PMD Sx		
50 %Un – Umax	20 %Ib – Imax	10 %In – Imax	45 – 55 Hz 55 – 65 Hz	±0.02 %

RMS Phase Current

PMD Types	Current Range	Min. Bandwidth (Harmonic)	Crest Factor	Class 0.2
PMD Sx	10 %In – Imax	45 Hz to 15 times rated frequency	2	±0.2 %
PMD Dx	20 %Ib – Imax			

Neutral Current Calculated (from Phase Currents)

PMD Types	Phase Current Range	Min. Bandwidth (Harmonic)	Crest Factor	Class 0.2
PMD Sx	10 %In – Imax	45 Hz to 15 times rated frequency	2	±0.2 % [1]
PMD Dx	20 %Ib – Imax			

1. Uncertainty is indicated as a percentage of the largest phase current value.

Neutral Current Measured (with a Sensor)

PMD Types	Neutral Current Range	Min. Bandwidth (Harmonic)	Crest Factor	Class 0.2
PMD Sx	10 %In – Imax	45 Hz to 15 times rated frequency	2	±0.2 %
PMD Dx	20 %Ib – Imax			

RMS Voltage

Voltage Range	Minimum Bandwidth (Harmonic)	Crest Factor	Class 0.2
Umin – Umax	45 Hz to 15 times rated frequency	1.5	±0.2 %

Power Factor

Voltage Range	Current Range		Power Factor	Class 0.5
	PMD Dx	PMD Sx		
50 %Un – Umax	20 %Ib – Imax	10 %In – Imax	0.5 inductive – 0.8 capacitive	±0.005 [1]

1. There is no unit for the value.

Appendix C Accuracy/Reliability

Accuracy

Measurement by Accura 3300E

Parameter		Accuracy	Measuring Range (Accuracy Guaranteed) / Minimum Measured Value	
Voltage	Line-to-neutral voltage	±0.2% Reading	35 – 277 V L-N / 5 V L-N	
	Line-to-line voltage	±0.2% Reading	60 – 480 V L-L / 9 V L-L	
Current		±0.2% Reading	1 %In – 200 %In [1] / 5 mA	Accura 3300E-5A
			5 %In – 200 %In [1] / 5 mA	Accura 3300E-1A
Power	Active	IEC 62053-22 Class 0.5S	1 %In – 200 %In	
	Reactive	IEC 62053-24 Class 0.5S	1 %In – 200 %In	
	Apparent	IEC 61557-12 Class 0.5	1 %In – 200 %In	
Energy	Active	IEC 62053-22 Class 0.5S	-	
	Reactive [2]	IEC 62053-24 Class 0.5S	-	
	Apparent [2]	IEC 61557-12 Class 0.5	-	
Demand	Power	IEC 62053-22 Class 0.5S	-	
	Current	±0.2% Reading	-	
Frequency		±10 mHz	42 – 65 Hz	
Power factor		±0.5% Full scale	1 %In – 200 %In	

1. The rated current In is 5 A/1 A.
2. The data are not displayed on the LCD screen of 3300E and are provided only via communication.

Accuracy of Power Quality Data

Item	Compliance	Condition	Accuracy
Voltage/Current phasor [1]	-	-	±0.5% Full scale
Voltage/Current unbalance [1]	IEC 61000-4-30, NEMA MG1	-	±0.5% Full scale
Voltage/Current THD [2]	-	Up to the 50th order	±0.5% Full scale
Current TDD [3]	-	Up to the 50th order	±0.5% Full scale
Crest factor [1]	-	-	±0.5% Full scale
K-factor [1]	-	-	±0.5% Full scale

1. The data are not displayed on the LCD screen of Accura 3300E and are provided only via communication.
2. THD(Total Harmonic Distortion), Voltage THD: $\frac{\sqrt{\sum_{k=2}^{50} V_k^2}}{V_1}$, Current THD: $\frac{\sqrt{\sum_{k=2}^{50} I_k^2}}{I_1}$
3. TDD(Total Demand Distortion). Current TDD: $\frac{\sqrt{\sum_{k=2}^{50} I_k^2}}{I_L}$

where the I_L value can be set to the peak demand current(default) or the rated current(only via communication) for calculation of TDD.

Reliability

Standards		Reference Value
EN 55011	Conducted emissions	150 kHz – 30 MHz
EN 55011	Conducted emissions (asymmetric mode)	-
EN 55011	Radiated emissions	30 MHz – 6 GHz
IEC 61000-3-2	Limits for harmonic current emissions	Equipment input current ≤16 A per phase
IEC 61000-3-3	Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems	Equipment with rated current ≤16 A per phase and not subject to conditional connection
IEC 61000-4-2	Electrostatic discharge immunity test	±4 kV/ ±8 kV contact/air
IEC 61000-4-3	Radiated, radio-frequency, electromagnetic field immunity test	(10 V/m) 80 MHz – 1,000 MHz (3 V/m) 1,400 MHz – 6,000 MHz
IEC 61000-4-4	Electrical fast transient/burst immunity test	±2 kV mains, ±1 kV IO signal/control
IEC 61000-4-5	Surge immunity test	±1 kV/±2 kV, line-to-line/line-to-earth ±2 kV mains, ±1 kV IO signal/control
IEC 61000-4-6	Immunity to conducted disturbances, induced by radio-frequency fields	3 V, 150 kHz – 80 MHz
IEC 61000-4-8	Power frequency magnetic field immunity test	30 A/m
IEC 61000-4-11	Voltage dips, short interruptions and voltage variations immunity test	Voltage dip, reduction 100/60/30 %, Period R1/12/30 Short interruptions, reduction 100 %, Period 300

Appendix D Ordering Information

Accura 3300E performs high accuracy measurements with ±0.2 % accuracy of reading for voltage and current and meets IEC 62053-22 class 0.5S standards for power and energy measurements. It also provides various power quality information including harmonics(up to the 50th order), Crest factor, K-factor, and unbalance, which are essential for power quality management of switchgear panels.

Device	Model	Option	Description
Digital Power Meter	Accura 3300E	-5A-AC220V/DC110V	Input current rating: 5 A Power supply: AC 100 – 240 V, DC 100 – 300 V
		-1A-AC220V/DC110V	Input current rating: 1 A Power supply: AC 100 – 240 V, DC 100 – 300 V
			Measures voltage/current/power
			Provides data on harmonics(up to the 50th order), unbalance, Crest factor, and K-factor
			Supports RS-485 communication
			Measures its temperature [1] using a sensor integrated into its MCU

1. It is not for fire monitoring, but for reference only.

Accura 3300E

User Guide

High Accuracy Digital Power Meter

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